

Estimating spatial firm productivity spillovers using UK microdata

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DECLARATION

I, Aloysius Lip, hereby declare that the work presented in this dissertation is my own original work. Where information has been derived from other sources, I confirm that this has been clearly and fully identified and acknowledged. No part of this dissertation contains material previously submitted to the examiners of this or any other university, or any material previously submitted for any other assessment.

I declare the use of GenAI tools in an assistative role (UCL Category 2) for this submission. The model Gemini 3.10 Pro (DeepMind, 2026) was used for limited roles of: supporting the particular process of reviewing and refactoring code (data analysis), and as a support tutor (economic model and empirical approach).

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Date: 2nd September, 2026

CLASSIFICATION

This piece of research is primarily (please indicate as appropriate):

- ☒ an empirical/econometric study
- ☐ the examination of a theoretical problem
- ☐ a critical analysis of a policy issue
- ☐ an analytical survey of empirical and/or theoretical literature

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ABSTRACT

I estimate the productivity spillover occurring between colocated firms in the United Kingdom. The productivity spillover, a spatial externality, underpins well-established economic phenomena around the development of urban clusters in the New Economic Geography literature, and is measured and heavily discussed in the evidence on the economic effects of urban density measured in the spatial econometrics literature, and underpinning regional growth policy.

I apply recently developed econometric techniques for valid identification when estimating spillover effects in a spatial group and network setting to measures of UK firm's Total Factor Productivity (TFP).

The key to valid identification rests in internal instruments from spatial *exogenous* effects in a static model, and from deep lags in a dynamic model: this overcomes Manski (1993)'s 'Reflection Problem' and the endogeneity of peer effects. It requires big data calculations of firm interactions, which I use modern data techniques to compute efficiently, preserving the advantages of Fame's high-observation balance sheet data for both its panel structure and its resulting high-significant estimates.

Across my instrumented specifications, I evaluate a spatial spillover of firm peers to any given target firm of an elasticity of (0.2), a significant proportion of GVA. These results imply that regional economic policy can have serious multiplier effects: more policy emphasis should be placed on raising the Total Factor Productivity of regional firms as its effects will spillover to its peers.

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1. INTRODUCTION

Estimating the strength of a spatial productivity spillover is crucial to economic discussions on growth, production, and sophisticated specialised fields such as the movement of households and firms across urban clusters. In particular, productivity spillovers can be discussed as a pure supply-side effect. It is notoriously tricky⁵ to isolate this effect from demand-side factors in general equilibrium from the movement and colocation of firms and people; even more so to isolate the pure spillover effect occurring through and on Total Factor Productivity (TFP) the residual to a firm's input factors.

The purpose of this paper is to demonstrate that modern advances in computing, theory, and econometric techniques mean that it is possible to make significant progress in this decades-long problem with credible estimates of the supply-side spatial spillover factors influencing TFP, even spanning from a stylised, classical model. In section 2, I give an overview of contributions from spatial and urban economics and economic geography, New Economic Geography (NEG) revisions and how the findings in this paper might substantiate the assumptions underlying NEG, and modern econometric literature.

In sections 3 and 4, I discuss the handling of data from Fame, a database of publicly-available, balance-sheet data from UK companies, unified and designed to be comprehensive for research. Fame offers the realisation of 100,000's of firms' data across over 1 million firm-yearly observations. This is a blessing for robust econometric analysis, but a curse for scalable, recreatable data handling to handle both the computation of millions of datapoints and their transformations, and safely manage its memory flow. In particular, I highlight two significant data challenges for this setting: the architecture design around spatial data, including calculating distance-to-distance peers and forming the spatial weighting matrix $\mathbf{W}$, and the iterated application and evaluation of high indices of $\mathbf{W}$, ex: $\mathbf{W}^2, \mathbf{W}^k, (\mathbb{I} - \mathbf{W})$ as both transformations in reduced form and to calculate valid instruments. The use of a high-performance, column vector-based data system, *DuckDB* was critical in this analysis for these purposes. All code for this analysis is available publicly⁶.

In section 5, I give the econometric analysis of a static model with spatial peer effects, conceptualisable as 'spatial lags' of the classical regressors in a 3-factor production function. This is an immediately gruelling application of Manski (1993)'s 'Reflection Problem', meaning the identification of *endogenous* regressors in a peer effects setting. I show that modern econometric theory on the 'Reflection Problem' in a network setting, as given in Bramoullé et al. (2009), cleanly and intuitively apply to this firm productivity spillover problem. I estimate a reduced-form peer effects model under competing exogeneity assumption and internal instrumental variable approaches. I undertake this for both a 'groups' model, which argues that the strength of a productivity spillover is the average of nearby neighbours, and a 'networks' model, which applies a individualised weight for each target-peer firm connection proportionate to the reciprocal of the physical distance between them. **(The distance model performs better under standard tests.)**

In section 6, I expand the use of internal instruments to use deep lags to identify exogenous variation in the otherwise-endogenous peer effects regressors. This builds on modern 'System GMM' methods from Blundell and Bond (1998). These are popular approaches to efficiently estimate coefficients from endogenous variables in a dynamic system exploiting the time variation of the panel structure. These models are likely the most robust given in this analysis. They suggest a more limited effect of peer-target productivity spillover. All specifications so a zero effect of a lagged TFP-spillover, and autoregressive term with a very narrow confidence interval, suggesting a robust result that if any productivity spillover does occur between firms, it happens in the contemporaneous period.

⁵Qualitative discussions in Krugman (1998), Fujita and Krugman (2004)

⁶<https://github.com/aloysiuslip/ucl-diss-ECON0079>

1.1. CONTRIBUTIONS FROM SPATIAL PRODUCTIVITY ANALYSIS TO POLITICAL ECONOMY

A revival of interest in regional, pro-second city policy has also taken place in the policies of the 2024 and 2026 governments in the United Kingdom. Regional economic policy declined strongly in the UK under the 1980s Thatcher government. . . This consensus was maintained throughout the 1990s-2000s New Labour period, where rapid economic growth known as ‘The Great Moderation’ provided no impetus for change. The consensus was broken in 2017, with the seeds of a regional-industrial strategy in 2017, emerging under the Conservative May government. Under the background of a UK-EU Referendum result (“Brexit”) which emphasised the sentiment of disaffection among ‘left-behind’ places across the UK which did not see the benefits of national aggregates of economic growth primarily concentrated in the UK’s capital, London. The national government launched ‘Local Industrial Strategies’ which sought to identify action to "boost productivity, earning power, and competitiveness" on a regional basis⁷. However, fewer of these strategies were fully developed and implemented by 2021. In 2026, the newly appointed Burnham government made the pledge for “good growth in every postcode”

1.2. ECONOMETRIC CONTRIBUTIONS

A key contribution by Manski (1993) is to argue that there is not just a direct, endogenous effect of the outcome of interest among peers, but also an exogenous effect of the characteristics of peers. However, Manski’s group model famously

but also a set of exogenous characteristics

It has been common to analyses of economic growth since Marshall (1920) to assume and qualitatively discuss some effect on firm-to-firm spillover of productivity. A productivity spillover underpins well-established economic phenomena such as the colocation of firms in urban clusters described in New Economic Geography literature such as Krugman (1991). It has been heavily analysed through industry and spatial density approaches in spatial and urban econometric literature such as Glaeser et al. (1992).

⁷<https://researchbriefings.files.parliament.uk/documents/CBP-7682/CBP-7682.pdf>

2. (LITERATURE)

2.1. AGGLOMERATION AND MARSHALL

My investigation is rooted in Marshall (1920)'s taxonomy, which identifies three channels to productivity spillovers: (1) input-output costs, (2) labour pooling, and (3) knowledge spillovers. There is no reason to think these are exhaustive. We might theorise that there may be competitive effects and cultural factors. Krugman (1991) also labels all these channels as 'centripetal' to an urban core; there are likewise 'centrifugal' factors that push firms away from the urban core, such as destructive (zero-sum) competition, congestion, and saturation costs. A further extension could develop models on these further channels and empirically test them. Furthermore, these productivity spillovers are incredibly salient to thinking about production structures.

In this investigation, I concentrate on (3) knowledge spillovers, tackling a difficult identification problem highlighted in Glaeser (2001). This is more relevant to the knowledge-based services economy that dominates the UK. Input-output costs and labour-pooling matter more for industrial manufacturing firms, especially the latter which was devised in response to the phenomenon emergence of industry towns.

Most agglomeration literature bundles all three channels together. Glaeser (2010) and Krugman (2010) identify how vital identifying knowledge spillovers are. That need to estimate them well only grows with the modern, knowledge economy of the UK. (Goldin, 2023) speculates that agglomeration effects have "increasing strength" attributable to the way in which "knowledge workers in particular become more productive when located near to one another. Information may be more available than ever, but the evidence suggests that it still tends to flow largely through in-person networks".

2.2. NEW ECONOMIC GEOGRAPHY

However, Krugman (1991) prefers not to assume this effect in their focus on the dynamic movement of firms. Instead, heterogeneous cluster policies, such as initial external differences in the number of workers or firms are frequently assumed, with 'forward and backward linkages' driving colocation endogenously from there. This paper helpfully tests the presence of spillovers, providing another centrifugal force directly incentivising firms to colocate.

2.3. MODERN APPROACHES

Specific approaches to investigating knowledge spillovers via innovation, instrumenting via patent publications comes from US evidence, including (Jaffe, 1993), (Berkes and Ruben, 2021) and more recently (Bloom, Schankerman and Van Reenen, 2013).

Knowledge spillovers have been a well-evidence factor of within-firm managerial-derived productivity gains. A field-starting MIT study from Allen (1977) observed an exponentially declining curve depicting the frequency of communication between two individuals against their physical distance in an office.

2.4. ECONOMETRIC LITERATURE

The result that the Manski linear-in-means (LMM) model is not identified is a famous one in peer effects economics, hindering the analysis of spatial and group dynamics between mutually influencing entities.

However, significant progress has been made of improving identification techniques, in particularly highlighting identification methods for more complex group and network structures in Bramoullé et al. (2009). Fortunately, as stated in the paper, many of these complex network structures are exactly what is common to empirical data, this investigation to the pattern and layout of firms interacting with estimation of production spillovers included.

A summary of the identification rules from Bramoullé et al. (2009) dependent on the structure of $\mathbf{W}$, a ‘weighting’ matrix of interactions between two firms in a matrix, is included below.

TABLE I
Spatial econometric identification rules for weighting matrix $\mathbf{W}$

Structure	Required condition	Identified	Instrument
<i>Group structure</i>			
Proposition 2	$\xi_0 - \delta_0\beta_1 \neq 0$ and $\zeta_0 - \delta_0\beta_2 \neq 0$		
Linear-in-Means ^a	Target is own peer	×	—
Leave-one-out ^b	Target excluded from peer mean	×	—
	Same group size		
Varying group sizes ^c	Target excluded from peer mean ≥ 2 group sizes	✓	$\mathbf{W}\mathbf{x}$
<i>Network structure</i>			
Proposition 1	$\mathbb{I}, \mathbf{W}, \mathbf{W}^2$ linearly independent $\beta\theta_0 + \xi_0 \neq 0$	✓	$\mathbf{W}^2\mathbf{x}$
Intransitive triad	$\exists i, j, k : W_{ij} > 0, W_{jk} > 0, W_{ik} = 0$	✓	$\mathbf{W}^2\mathbf{x}$
Transitive networks	$\mathbf{W}^2 \neq 0$	✓	$\mathbf{W}^2\mathbf{x}$
Deep network	$s_d \geq 2$ network diameter	✓	$\mathbf{W}^2\mathbf{x}$
Proposition 3	$\mathbf{W}^2 \neq 0; \beta\theta_0 + \xi_0 \neq 0$	✓	$\mathbf{W}^2\mathbf{x}$
$\mathbf{W}^2 = 0$	$\alpha \neq 0$	✓	$\mathbf{W}^2\mathbf{x}$
<i>Adding correlated effects</i>			
Proposition 4	$\mathbb{I}, \mathbf{W}, \mathbf{W}^2, \mathbf{W}^3$ linearly independent $\beta\theta_0 + \xi_0 \neq 0$	✓	$(\mathbb{I} - \mathbf{W})\mathbf{W}^2\mathbf{x}$
Group structure	$s_g \geq 3$ group sizes	✓	$(\mathbb{I} - \mathbf{H})\mathbf{W}^2\mathbf{x}$
Network structure	$s_d \geq 3$ network diameter	✓	$(\mathbb{I} - \mathbf{H})\mathbf{W}^2\mathbf{x}$
Proposition 5	$\mathbf{W}^3 = \lambda_0\mathbb{I} + \lambda_1\mathbf{W} + \lambda_2\mathbf{W}^2$ $\text{rank}(\mathbb{I} - \mathbf{W}) < n - 1$ $2\lambda_0 + \lambda_1 + 1 \neq 0$	✓	$(\mathbb{I} - \mathbf{H})\mathbf{W}^2\mathbf{x}$

Adapted from Bramoullé et al., 2009. (a) Manski, 1993, in this model $\mathbf{W}^2 = \mathbf{W}$ for linear dependence. (b) Moffitt and Caspi, 2001, in this model, $\mathbf{W}^2 = \frac{1}{s-1}\mathbb{I} + \frac{s-2}{s-1}\mathbf{W}$ for linear dependence. (c) fei Lee, 2007.

3. FINANCIAL DATA

3.1. FAME

I have used data from Fame for this investigation. Fame is a comprehensive set of financial datapoints from UK firms with useful variables for this analysis, in particular, number of employees and balance sheet data from which analysis-relevant aggregates can be derived. Fame is provided by Moody's Analytics, an international credit agency, and collects its data from UK firms via survey⁸.

Its international counterpart, Orbis, offers a similar service for international counterparts. This analysis could therefore be extended to offer insights comparing the spillover effects of UK firms to their international counterparts. For this investigation, the UK dataset is comprehensive and the most policy-relevant for this investigation.

Raw data was accessed from Fame in February 2025 by Professor Lars Nesheim for Irastorza-Fadrique et al. (2026) and reused for this project. Raw data is provided on a series of a financial balance sheet indicators, including characteristics that are fixed for each firm, bucketed into *a1_ID* and *a5_misc* indicator tables, and annual data published on a calendar-year basis, bucketed into *a2_key_finance*, *a3_assets*, and *a4_profits* tables.

3.2. (INGESTION AND CLEANING)

I designed a series of Python scripts, organised in Jupyter notebooks to ingest this data and manipulate this data. Data was stored in *DuckDB*, an open-source, MIT-licensed relational (SQL) database architecture⁹. It operates locally, meaning minimal lag from issuance to execution time provided the machine of operation is sufficiently high-compute. It stores data in column vectors, which is a significant advantage to manipulating whole variables, for instance to apply Matrix transformations as is required to calculate spatially-weighted distances. Any SQL commands issued from the Python are compiled down to C, making execution for column transformation operations extremely fast, even for joining tables of several 100s of millions of rows, as is calculated in this analysis for target-peer spatial distances. I wrote database interaction commands using *Ibis*¹⁰, an Apache-2.0 licensed Object-Relational Mapping framework (ORM) which allows SQL commands to be written natively in Python, minimising the likelihood of errors when issuing queries loops or changing variable names. This architecture choice is extremely well-suited for mass data econometric data analysis and critical to solving the compute limitations of iterative large-*n* dataset manipulations required to calculate the valid instruments identified in section 5.1.

(...)

All data is deflated to real terms, using 2024 as the base year. I used the GDP deflator for all balance sheet variables, except for *renumeration_employees* and its composites for which I used CPIH, as these are income-related aggregates with real spending power best captured by household purchasing baskets¹¹.

All balance sheet data are in thousands ('000s) of GBP (£). As we estimate log relationships exclusively, this has no effect on parameter interpretation.

3.3. DERIVING PANEL AGGREGATES

Labour input is taken from the number of employees reported on the FAME balance sheet per firm. This dataset, like all others, does not include hours worked per employee on a firm-level that would offer a more granular productivity (GVA per hour worked) measures. An estimation in this setting would likely

⁸Analytics (2025)

⁹Raasveldt and Muehleisen (2026)

¹⁰Authors (2026)

¹¹for National Statistics (2026)

have to impute the measure from industry aggregates, which would likely confound the true firm-level signal.

Capital is Fame’s *fixed_total* variable, which represents the sum of tangibles, intangibles, and other investments within the balance sheet, namely an extremely comprehensive measure of the firm’s fixed capital. The average capital-output ratio is 6x within FAME when filtering for positive values Table (II), which is moderately higher than macroeconomic aggregates typically suggest likely due to accounting standard differences.

TABLE II
Capital-output ratios in FAME corpus of firms

Variable	Description	Capital-output ratios
<i>tangibles</i>		3.53
<i>fixed_total</i>	tangibles + intangibles + inv_other	6.00
<i>total_assets</i>	fixed_total + current_assets	11.56

GVA is calculated as the sum of *ebitda* and *remuneration_employees*. *remuneration_employees* is the exact sum of *wages*, *social_security_costs*, and *pensions_costs* in the balance sheet. *ebitda* is in turn *turnover* minus intermediate inputs and is given in the Fame balance sheet for the vast majority of panel observations in our dataset. It is also possible to reconstruct *ebitda* via the sum of *profit_loss_pretax*, *interest_paid*, and *depreciation*, providing an alternative measure of GVA. We can expect the difference between these two to be accounting differences¹² and additionally note that the alternate GVA measure also reduces panel observations from 1,085,520 to 618,588. For these reasons, I do not report TFP calculations with this second GVA measure which are regardless near-identical, though I resultingly refer to GVA throughout this paper as *gva_1*.

To help guide the choice of capital and output (GVA) measure, I conduct a quick benchmark 2-factor production estimation to help validate the choice of aggregates against known measurements.

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + u_{i,t} \quad (1)$$

This specification is not valid OLS specification by the structural model I give later in 5.1. $u_{i,t}$ is a non-structural error-residual that is correlated with $k_{i,t}$, $l_{i,t}$. The estimation is a simple panel approach with time and fixed effects, and a constant. Regressors are given in log terms to approximate a structural production function similar to 4.

Coefficients derived in Table (III) give a labour income share, $\hat{\beta}_2$, exactly in line with ONS analysis of UK industry-level data over the same time period, particularly *corporate gross labour share* which is constructed from a similar sample to the coverage in Fame¹³.

Surprisingly, running the regression with *fixed_total* as the capital measure instead of *total_assets* almost entirely eliminates the capital share of income, or explanatory power of capital on variation in GVA. I leave it for future analysis why this dataset yields this phenomenon and for the rest of the analysis, I use *total_assets* as the primary capital measure.

As a demonstration, I include specifications which omit fixed effects. The capital share of income jumps on a specification with only time fixed effects, which is to be expected (explain).

¹²An OLS regression between the two, discarding 5% outliers, has a coefficient of 0.91 with a 0.000 p-value. This means *gva2* is slightly lower than *gva1*, with that impact mostly driven by top-end outliers.

¹³for National Statistics (2024)

TABLE III
Baseline 2-factor TFP estimation: varying fixed effects

	(1.) <i>gva</i> ₁	(2.) <i>gva</i> ₁	(3.) <i>gva</i> ₁	(4.) <i>gva</i> ₁
constant	2.995 (0.036)***	3.036 (0.031)***	1.970 (0.010)***	1.973 (0.010)***
<i>ln(employees)</i>	0.617 (0.005)***	0.614 (0.004)***	0.561 (0.002)***	0.561 (0.002)***
<i>ln(total_assets)</i>	0.286 (0.005)***	0.284 (0.004)***	0.421 (0.002)***	0.421 (0.002)***
<i>i</i> fixed effects	✓	✓		
<i>t</i> fixed effects	✓		✓	
Observations	1,045,164	1,045,164	1,045,164	1,045,164
R ² (excl. F.E.)	0.3313	0.3453	0.8094	0.8096
R ² (Overall)	0.7912	0.7900	0.8096	0.8096

TABLE IV
... varying parameter measures

	(5.) <i>gva</i> ₁	(6.) <i>gva</i> ₂	(7.) <i>gva</i> ₁	(8.) <i>gva</i> ₁
constant	2.995 (0.036)***	3.288 (0.056)***	4.751 (0.018)***	5.268 (0.070)***
<i>ln(employees)</i>	0.617 (0.005)***	0.611 (0.007)***	0.720 (0.004)***	0.628 (0.015)***
<i>ln(total_assets)</i>	0.286 (0.005)***	0.265 (0.007)***		
<i>ln(fixed_total)</i>			0.065 (0.002)***	
<i>ln(tangibles)</i>				0.090 (0.006)***
<i>ln(intangibles)</i>				0.025 (0.002)***
<i>ln(investments_other)</i>				0.007 (0.001)***
<i>i</i> fixed effects	✓	✓	✓	✓
<i>t</i> fixed effects	✓	✓	✓	✓
Observations	1,045,164	603,586	1,006,904	152,444
R ² (excl. F.E.)	0.3313	0.2764	0.2830	0.2699
R ² (Overall)	0.7912	0.7390	0.6751	0.7403

All parameters reported with standard errors underneath. Significant estimates against an $H_0 : \beta = 0$ are indicated with $p < 0.1$ (*), $p < 0.05$ (**), and $p < 0.01$ (***).

4. LOCATION DATA AND INTERACTIONS

Throughout this analysis, I use terminology of a productivity spillover occurring from a group or average of *peer* firms onto a *target* firm. The influence of every firm j on the vector of target firms i can be summarised in an $(n \times n)$ interactions matrix, $\mathbf{W}$.

(This can be explained better: in a table with 4 columns) The interactions relationship is irreflexive. For groups it is transitive: if firms A and B share a spatial group, such as a travel-to-work area (*ttwa*), first-half of a UK postcode (*pc4*), or full postcode (*pc8*), then a firm C that is a peer of firm B must also be of firm A. For continuous distance measures, this is non-transitive. Firm B, a peer of firm A within its 5km radius, can clearly have its own peers that are within its own 5km radius that are not shared with A. Every firm is then its own overlapping group, or equivalently, a sparse, rather than block diagonal, firm-to-firm matrix¹⁴.

(Cleaning address data).

4.1. LOCATION DATA LIMITATIONS

Fame only reports each company's last-known registered and trading address, at the time of compilation of publication of its annual vintage (cite). As a result, final location data is only available at the firm level. We can interpret each firm as being static over the entire 20-year sample, with no movers. This seems very unlikely (expand). Instead, we must interpret the constructing address, and therefore firm-to-firm distance measures as a signal of firm pairs' true distance polluted by movers. This has 2 econometric consequences.

Firstly, firm fixed effects in all spatial specifications become extremely important. For all movers within the sample period, their firm location decision is endogenous to the subsequent productivity spillovers they experience and exert, for reasons discussed in Section 2 on New Economic Geography. (More on firm FEs) There is no solution to this for moves occurring out-of-sample, for instance, just before 2009.

Secondly, I acknowledge some attenuation to the productivity spillover estimate Peer firms who were close to the target firm during the firm-year observation who subsequently moved away within-sample will have an extreme distance recorded, and so be excluded from tagged address groups and truncated distance calculations. Peer firms who are recorded as close to the target firm as they subsequently moved to the location (More on direction of bias/positive selection?)

4.2. LOCATION GROUP CLASSIFICATION

I first consider a model where each target has peers which share a spatial identifier, meaning they belong to the same travel-to-work-area (*ttwa*) using the 2011 definitions¹⁵, the first half of a UK postcode *pc4*, and the full UK postcode *pc8*.

The structure of the weighting matrix $\mathbf{W}$ is key to identification. W is an $n \times n$ matrix of weights where each column j denotes a weight w_{ij} applied on the variable of peer firm j , summing across rows i to reach a weighted aggregate regressor with an estimated effect on target firm i .

A first property of W is that its diagonal elements are all 0: a firm's TFP aggregate cannot be part of a weighted sum to an effect on its own TFP or GVA. This is common to peer effects literature such as Moffitt and Caspi (2001) and discussed in Bramoullé et al. (2009).

Group classifiers impose an otherwise block diagonal structure on $\mathbf{W}$. The resulting $n \times n$ matrix has off-diagonal elements as the number of group members $n_g - 1$, which excludes the target firm i . The

¹⁴I do actually impose a block diagonal structure for reasons of computation as explained subsequently

¹⁵(Review ONS bulletin on TTWAs data source and describe clearly)

structure is given below¹⁶:

$$w_{ij} \in \mathbf{W}_g = \begin{cases} 0, & \text{if } i = j \\ \frac{1}{n_g-1}, & \text{else if } (i, j) \in g \\ 0, & \text{otherwise} \end{cases}$$

$$\mathbf{\Gamma}_{s_1} \in \mathbf{W}_g = \{i, j \in g\}, s_1 : n_g$$

$$\mathbf{W}_g = \begin{pmatrix} \mathbf{\Gamma}_{s_1} & 0 & \dots \\ 0 & \mathbf{\Gamma}_{s_2} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}_{(n \times n)}$$

An example such matrix is $\mathbf{W}_{g^*}$, a (7×7) set of interactions between firms which have group classifiers of size $s_1 = 4$ and $s_2 = 3$ only.

FIGURE I
Illustration of group matrix $\mathbf{W}$

$$\mathbf{W}_{g \sim I} = \begin{pmatrix} 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$$

Weighting matrix $\mathbf{W}$ illustration for a network of 7 interacting firms in a group $pc8$. Of these, firms 1-4 share their postcode, as do firms 5-7.

I can extend this to more complicated set of group rules. I calculate non-overlapping group weights $\mathbf{W}_g$ to $\mathbf{W}_{g1}, \mathbf{W}_{g2}, \mathbf{W}_{g3}$ at once, which represent the successive concentric rings of ($pc8$) (core), $pc4$, and $ttwa$. The weighting rules for these multiple, non-overlapping groups is to zero the i, j interaction in $\mathbf{W}_{gk}$ if a prior matrix in the series, $\mathbf{W}_{gl}$, $l < k$ has a non-zero entry in that cell. I illustrate an example in Figure II. These matrices are orthogonal.

$$w_{ij} \in \mathbf{W}_{gk} = \begin{cases} 0, & \text{if } i = j \\ 0, & \text{if } w_{ij,l} \neq 0, \exists l < k \\ \frac{1}{n_g-1}, & \text{else if } (i, j) \in g \\ 0, & \text{otherwise} \end{cases}$$

Firms have been labelled using their location features to symmetric, transitive, non-overlapping groups by $ttwa$ (largest) to lat_lon5 (smallest), with the latter meaning two firm entries shared a latitude-longitude identifier to 5 decimal places. Reviewing the summary statistics, I opted for full postcodes, $pc8$ as the group identifier for Table ??1.. This was mostly driven by full data availability, and the consistency of postcode definitions with the larger group $pc4$. Deriving a subgroup by latitude-longitude match within the postcode seemed fruitless, as in many cases it would identify the exact same firms, a move from the average number of peers from 3.1 to 2.9, and was a similar number of groups, with non-cohesive definitions.

¹⁶Adapted from Bramoullé et al. (2009)

FIGURE II
Illustration of multiple, non-overlapping groups

$$\mathbf{W}_{g \sim II} \sim \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & 0 & 0 \\ \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & 0 & 0 \\ \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & 0 & 0 \end{pmatrix}$$

Non-overlapping group illustration for a network of 8 interacting firms in a *ttwa*. Of these, firms 1-5 share the first half of a postcode - are in a *pc4* group, as are firms 6-8. Within those, firms 1-3 share a full postcode, as do firms 7-8. Firms 4-6 are isolated in their *pc8* groups.

TABLE V
Summary statistics across location groups

Property	Number of groups	Unclassified	Unclassified (%)	Avg. peers per group
<i>ttwa</i>	228	567	0.4%	668.3
<i>pc4</i>	2,705	0	0.0%	56.3
<i>pc8</i>	49,963	0	0.0%	3.1
<i>lat_lon3</i>	48,540	507	0.3%	3.1
<i>lat_lon4</i>	53,193	507	0.3%	2.9
<i>lat_lon5</i>	53,246	507	0.3%	2.9

4.3. (LOCATION DISTANCE CALCULATIONS)

4.4. ITERATIVE WEIGHTING APPLICATIONS

Intuitively, I take the product of these weighting matrices $\mathbf{W}$ to the aggregate variables vectors *gval*, capital *total_assets*, and labour *employees*. As will be shown in section 5 Static models, the procedure to generate valid instrumental variables also requires me to apply multiples of this transformation matrix, exponents $\mathbf{W}^2$, $\mathbf{W}^3$, $\mathbf{W}^4$, and for interacting group matrices, $\mathbf{W}_{g1}\mathbf{W}_{g2}$, $\mathbf{W}_{g2}\mathbf{W}_{g3}$ and so on to the aggregates.

As a data procedure, I calculate these for my 3 aggregate variable vectors of interest. As given below in Table VI, this requires a total of 51 compound weighting matrices, or columns of a database, to be calculated and generated. Of course, these column vectors each contain a stacked *firm (i)-year (t)* observation, giving an additional matrix of over 1 million entries.

Matrix products, including taking high indices, of the interactions between 1,52,379 firms is not computationally difficult, especially with sophisticated algorithms to handle $\mathbf{W}$'s block diagonal structure. However, storing the calculation in temporary memory is beyond the capacity of any standard machine available as a research tool.

Instead, I use a computational approach that avoids calculating the $\mathbf{W}^2$, $\mathbf{W}^3$... matrix separately, but rather iteratively applies the last pre-multiplied matrix to the aggregate variable, and then pre-multiplies the next matrix along to the result. Using memory-efficient *duckdb*, the procedure is as follows: (itemize)

4.5. EVALUATING THE LINEAR INDEPENDENCE OF $\mathbf{W}$ MATRICES

As will be discussed more heavily in the theory of 5.1, the key condition of these $\mathbf{W}$ matrices that will permit valid instrumentation of endogenous variables is to check the linear independence of $\mathbf{I}$, $\mathbf{W}$, $\mathbf{W}^2$, and later $\mathbf{W}^3$. I can trivially say this is true, meeting the alternative sufficient conditions in I of ≥ 3 group

TABLE VI
Variable names from transforming $\mathbf{y}_t$ and $\mathbf{x}_t$ regressors via weighting matrix $\mathbf{W}$

		endogenous	exogenous	
	Matrix product	gva	capital	labour
<i>Groups</i>				
regressors	$\mathbf{W}_{g1}$	wg1_y	wg1_k	wg1_l
	$\mathbf{W}_{g2}$	wg2_y	wg2_k	wg2_l
	$\mathbf{W}_{g3}$	wg3_y	wg3_k	wg3_l
instruments	$\mathbf{W}_{g1}^2$		w2g1_k	w2g1_l
	$\mathbf{W}_{g1}\mathbf{W}_{g2}$		wg1wg2_k	wg1wg2_l
	$\mathbf{W}_{g2}\mathbf{W}_{g3}$		wg2wg3_k	wg2wg3_l
	$\mathbf{W}_{g3}\mathbf{W}_{g1}$		wg3wg1_k	wg3wg1_l
	$\mathbf{W}_{g2}\mathbf{W}_{g1}$		wg2wg1_k	wg2wg1_l
	$\mathbf{W}_{g3}\mathbf{W}_{g2}$		wg3wg2_k	wg3wg2_l
	$\mathbf{W}_{g3}\mathbf{W}_{g1}$		wg3wg1_k	wg3wg1_l
local network FE	$(\mathbb{I} - \mathbf{W}_{g1})\mathbf{W}_{g1}^2$		(i-wg1)w2g1_k	(i-wg1)w2g1_l
global network FE	$(\mathbb{I} - \mathbf{V}_{g1})\mathbf{W}_{g1}^2$		(i-vg1)w2g1_k	(i-vg1)w2g1_l
<i>Network</i>				
regressors	$\mathbf{W}_{d1}$	wd1_y	wd1_k	wd1_l
	$\mathbf{W}_{d2}$	wd2_y	wd2_k	wd2_l
	$\mathbf{W}_{d3}$	wd3_y	wd3_k	wd3_l
instruments	$\mathbf{W}_{d1}^2$		w2d1_k	w2d1_l
	$\mathbf{W}_{d2}^2$		w2d2_k	w2d2_l
	$\mathbf{W}_{d3}^2$		w2d3_k	w2d3_l
	$\mathbf{W}_{d1}^3$		w3d1_k	w3d1_l
	$\mathbf{W}_{d1}^4$		w3d2_k	w3d2_l
local network FE	$(\mathbb{I} - \mathbf{W}_{d1})\mathbf{W}_{d1}^4$		w3d3_k	w3d3_l
global network FE	$(\mathbb{I} - \mathbf{V}_{d1})\mathbf{W}_{d1}^4$		w4d1_k	w4d1_l

Total: 51 new variables to be created, for a total of 53 columns in a linked table.

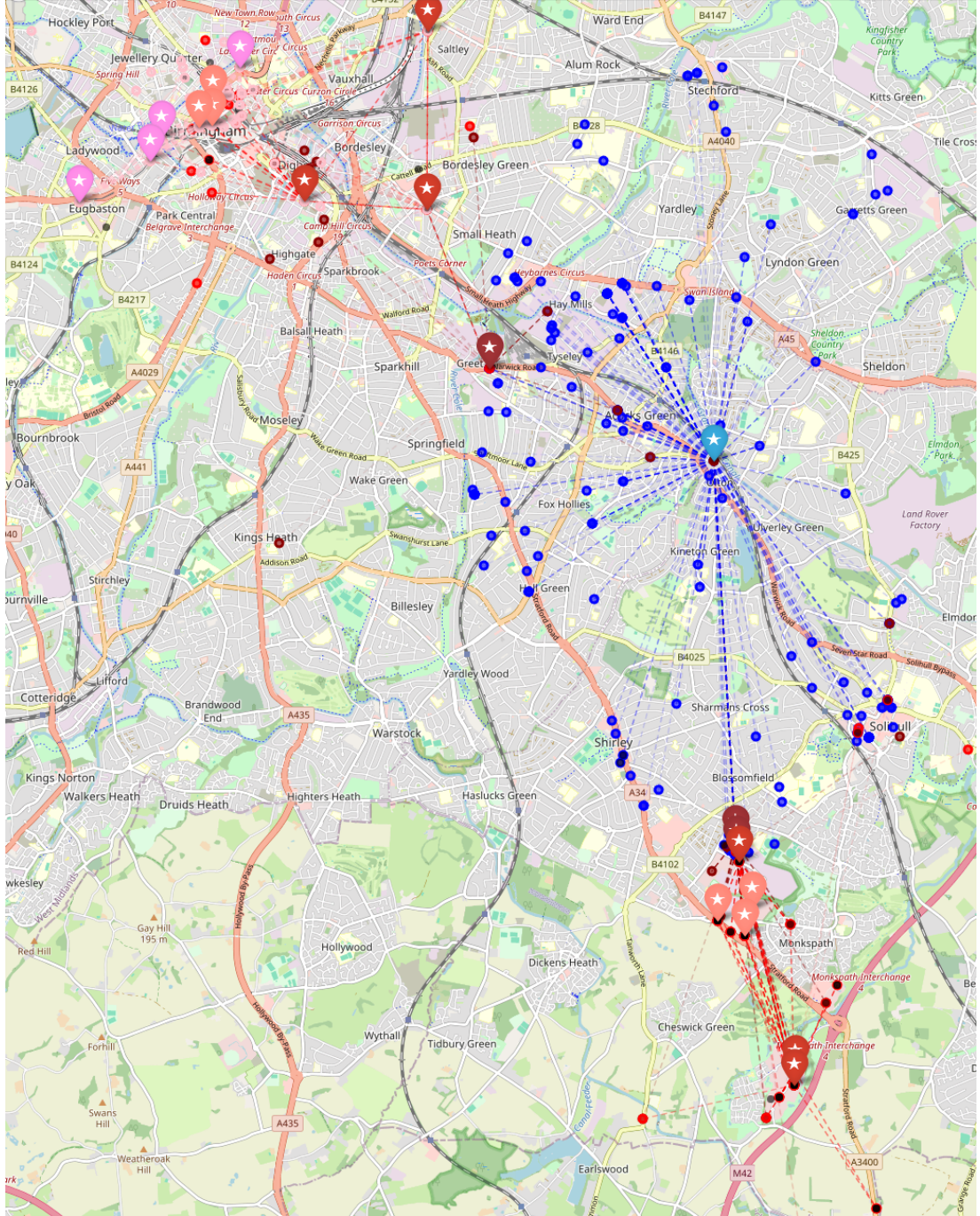
sizes for the group $\mathbf{W}$ matrices. (Histogram of group sizes?)

My distance-decay weights, $\mathbf{W}_{d1}$, $\mathbf{W}_{d2}$, $\mathbf{W}_{d3}$ meet the strong alternative $s_d \geq 3$ condition, where s_d is the *diameter* of the network, the maximal number of links connecting any target i with another firm through a series of peers $j, k, \dots$. In the analysis of this set of firms with distance radius limits, the *diameter* is large, potentially spanning the full spatial diameter of a *ttwa*. It is this mechanism design provides another justification, alongside computation, for limiting the max length of a target-firm peer relation to 5km: it vastly increases the probability of ‘overlapping groups’, i.e. a high diameter network within a ‘itxttwa’. The overlapping group network structure lends itself nicely to illustrations. Figure III selects a random firm i from my sample, which happens to be in Acocks Green, Birmingham and draws the distances used to calculate metric $d1$ between this target and its peers. In typical network diagram fashion, a network with a considerable *diameter* is created by taking a subset of i ’s peers, then successively a set of their peers, then successively some of those and so on. 4 iterations in this diagram fails to clearly hit the *ittwa* boundary, on the south side, with more argument for this behaviour on the north side where peer nodes randomly selected may be starting to bunch at an ‘invisible’ border of the *ttwa* boundary.

As additional credibility, I discuss strategies for evaluating the linear independence of $\mathbb{I}$, $\mathbf{W}$, $\mathbf{W}^2$ (...)

FIGURE III

Illustration of overlapping groups from a target firm to its whole network through successive peers



(blue pin): target firm i with a full set of its (blue dot) peers j , distances d_1 metric drawn in.

(dark red pins): sample (limit 5) of j peers, then calculated as their own targets

(red pins): sample (limit 5) of k peers of firms j , some of which are also shared as a peer of i .

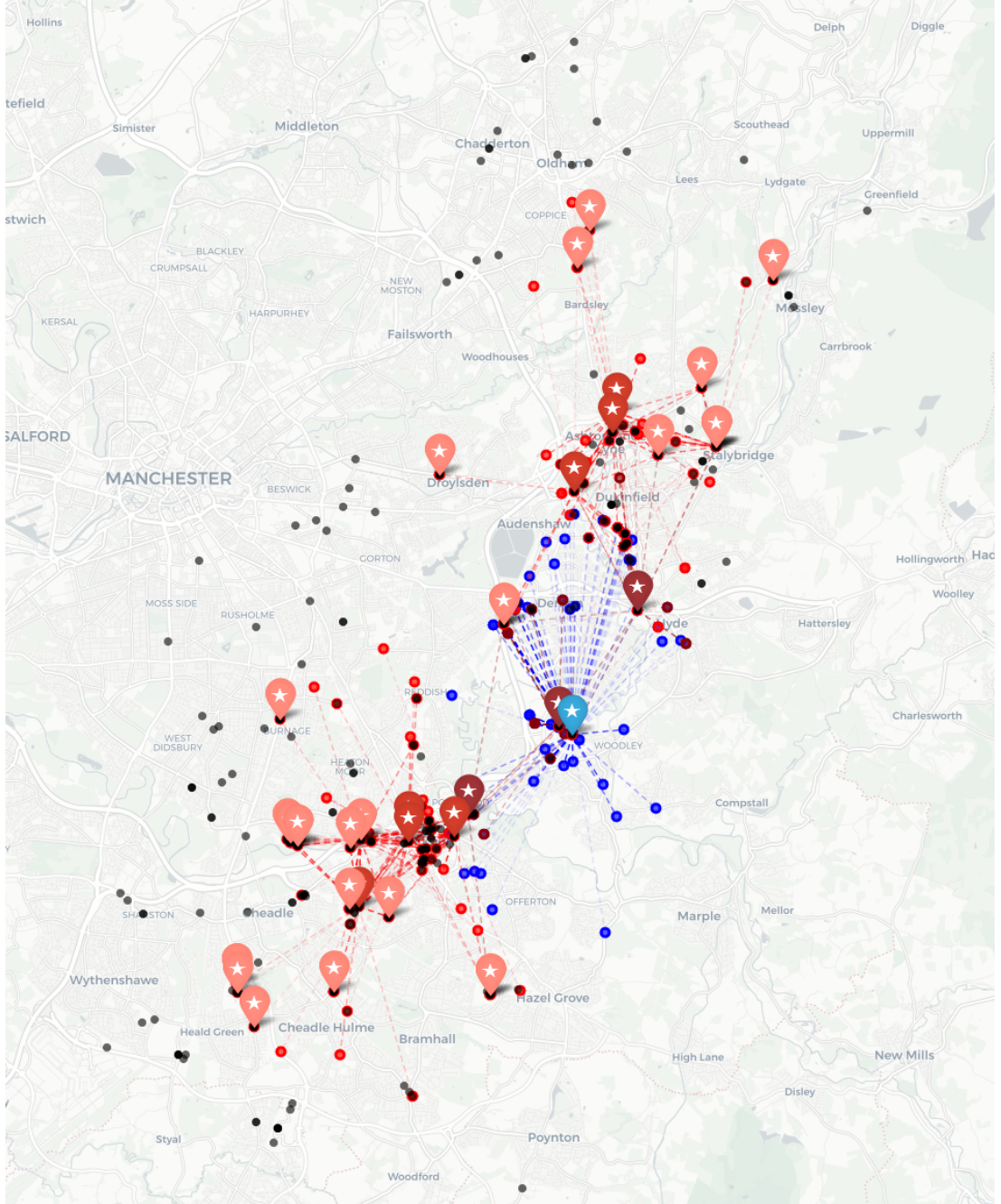
(light red pins): sample (limit 5) of m peers of firms k , few of which are also shared as a peer of i .

(light red dots): set of n peers of firms m , few of these are peers of i, j , probabilistically creating a network diameter of $s_d \geq 3$.

Plotting software from Folium (2012). Map tiles from OpenStreetMap (2017).

FIGURE IV

Illustration (2) of overlapping groups from a target firm to its whole network through successive peers



(blue pin): target firm i with a full set of its (blue dot) peers j , distances d_1 metric drawn in.

(dark red pins): sample (limit 5) of j peers, then calculated as their own targets

(red pins): sample (limit 5) of k peers of firms j , some of which are also shared as a peer of i .

(light red pins): sample (limit 5) of m peers of firms k , few of which are also shared as a peer of i .

(light red dots): set of n peers of firms m , few of these are peers of i, j , probabilistically creating a network diameter of $s_d \geq 3$.

Plotting software from Folium (2012). Map tiles from CARTO (2026).

5. STATIC MODELS

I use a stylised 3-factor production model throughout this analysis. Y , A , K , and L are each firm-year observation's GVA, TFP, labour, and capital in levels. I use lower-case variables to indicate logged terms and bolded variables to indicate i -stacked vectors.

Log TFP, denoted by $z_{i,t}$, is the residual after the own-firm contribution of the two classical inputs. I decompose it to a constant across all observations, firm and time fixed effects and, most crucially for this investigation, a vector of the weighted sum of all peer 3-factor production factors with the dot product against the vector of their respective spillover coefficients θ_0 :

$$\mathbf{z}_t^{spill} = \left(\sum_{j \neq i} w_{ij} z_{j,t} \quad \sum_{j \neq i} w_{ij} k_{j,t} \quad \sum_{j \neq i} w_{ij} l_{j,t} \right) \quad (2)$$

$$\theta_0 = \begin{pmatrix} \delta_0 & \xi_0 & \zeta_0 \end{pmatrix}' \quad (3)$$

Different static weights, w_{ij} , are explored throughout this section. The objective is to estimate the vector of structural coefficients θ_0 of spillover from peers to any given target firm.

$$Y_{i,t} = A_{i,t} \cdot K_{i,t}^{\beta_1} \cdot L_{i,t}^{\beta_2} \quad (4)$$

$$z_{i,t} = \ln(A_{i,t}) = y_{i,t} - \beta_1 k_{i,t} - \beta_2 l_{i,t} \quad (5)$$

$$= \alpha_i + \gamma_t + \beta_0 + \mathbf{z}_t^{spill} \theta_0 + \varepsilon_{i,t} \quad (6)$$

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \xi_0 \sum_{j \neq i} w_{ij} k_{j,t} + \zeta_0 \sum_{j \neq i} w_{ij} l_{j,t} + \varepsilon_{i,t} \quad (7)$$

$$\begin{aligned} &= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \varepsilon_{i,t} \\ &\quad + \delta_0 \sum_{j \neq i} w_{ij} \cdot (y_{j,t} - \beta_1 k_{j,t} - \beta_2 l_{j,t}) + \xi_0 \sum_{j \neq i} w_{ij} k_{j,t} + \zeta_0 \sum_{j \neq i} w_{ij} l_{j,t} \\ &= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \varepsilon_{i,t} \\ &\quad + \delta_0 \sum_{j \neq i} w_{ij} y_{j,t} + (\xi_0 - \delta_0 \beta_1) \cdot \sum_{j \neq i} w_{ij} k_{j,t} + (\zeta_0 - \delta_0 \beta_2) \cdot \sum_{j \neq i} w_{ij} l_{j,t} \end{aligned} \quad (8)$$

Equation (8) is a linear peer effects specification as given by Manski (1993). It is helpful to borrow the terminology from this literature think of $\begin{pmatrix} k_{i,t} & l_{i,t} \end{pmatrix} \in x_{i,t}$ as *exogenous* characteristics of firm i causally affecting its outcome $y_{i,t}$. There is an *endogenous* effect directly from peer outcomes $y_{j,t}$ with coefficient δ_0 . Peers also have an *exogenous* characteristics $x_{i,t}$, which have a direct impact on on the target's productivity $x_{i,t}$ via ξ_0 and ζ_0 . For this section of models, I assume there are no *correlated* effects¹⁷. I use this peer effects terminology throughout this paper italicising *endogenous*, *exogenous*, and *correlated*, noting to the reader the confusion this may cause with the more general and related concepts of endogeneity and exogeneity. The use of fixed effects has no effect on econometric interpretation of the peer effects model.

It is important to note that I do not impose restrictions on θ_0 . I allow each of the 3 factors of production to independently exert a spillover effect in general form. This allows this analysis' investigation as to which of the production factors matters more for productivity spillovers, and in what ratio, conceptually linked to the discussion in 2.1 about identifying the channels through which productivity spillovers might occur. Through this 3-factor production model, it is hard to link each parameter in θ_0 to a Marshallian channel, but broadly we might consider ζ_0 , the coefficient on an increase in a peer firm's labour inputs affecting a target's TFP to resemble Marshall (1920)'s labour pooling, with some interaction to knowledge spillovers, and δ_0 , the coefficient from a peer firm's TFP to target TFP to be the remainder of knowledge spillover effects. A TFP spillover occurring from peer firm's capital ξ_0 would be

¹⁷Though this would be achieved via network fixed effects

a novel channel to Marshall (1920)'s discussion and could reflect the idea of more capital-intensive peers. Regardless, it is strongly justifiable that we would expect the coefficient on each peer production factor to spillover to a target firm's production heterogeneously, and not be proportionate to the contribution of these factors to their own-firm GVA. I have no *a priori* arguments for the sign of the θ_0 coefficients and their relative magnitudes compared to their contributions to own-firm GVA in β_1, β_2 .

$$y_{i,t}^{(y)} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} y_{j,t} + \varepsilon_{i,t} \quad (9)$$

$$\begin{aligned} y_{i,t}^{(z)} &= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \\ &= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} y_{j,t} - \delta_0 \beta_1 \sum_{j \neq i} w_{ij} k_{j,t} - \delta_0 \beta_2 \sum_{j \neq i} w_{ij} l_{j,t} \end{aligned} \quad (10)$$

Two alternative specifications may be proposed: equation (9) modifies (7) to argue that it is the whole of GVA which matters the peer-target spillover, not each component heterogeneously. It is trivial to see that this implies the coefficient on the *exogenous effect* is zero, such that $\xi - \delta_0 \beta_1 = \zeta - \delta_0 \beta_2 = 0$. This has the interpretation alluded to above, that the peer effect of any classical factor, capital and labour, is exactly in the ratio to its own factor income, which is a strong restriction to impose. We might expect this to be a nested specification of (8), that we can merely Wald-test the joint hypothesis as given above, but actually this specification cannot be identified due to the endogeneity of $y_{j,t}$, without strict assumptions on the orthogonality of peer errors.

The alternative in equation (10) argues that only the TFP component of production matters for the spillover to influence a target firm's TFP. This is also a restriction on the more general (7) such that $\xi_0 = \zeta_0 = 0$. On first glance, it may appear that this model follows the structure of the peer effect class of models too and is in fact over-identified, with both the coefficients of the peer classical factors and $\sum_{j \neq i} w_{ij} y_{j,t}$ identifying δ_0 . However, because of the specific set-up of the reflection problem where $w_{ij} y_{j,t}$ is endogenous and we instrument it through $k_{j,t}, l_{j,t}$ higher-order spatial lags, this model is actually non-identified. I demonstrate this in Appendix 1: identification under differing spillover assumptions.

While the classical production function (7) is limited to constant returns to scale, $\beta_1 + \beta_2 \leq 0$, a standard assumption across the growth and firm production literature, no such restrictions for the firm occur with the endogenous decomposition of TFP. A restriction on logged production (7) collectively on β_1, β_2 , and θ_0 is difficult to justify. As with other external effects, there is no reason to believe that market factor prices and the firm decision-rule for spatial allocation in view of these spillover effects, $w_{ij} z_{j,t}$, is socially optimal. A peer firm j with extremely high and growing components of GVA has little interest in siting themselves in the centre of a cluster of firms that are struggling with TFP growth, yet the external impacts of that firm would be significant.

5.1. STATIC PANEL IDENTIFICATION

Consistency requires the plausible exogeneity of regressors used in structural equation (8) from the error term $\varepsilon_{i,t}$. Capital and labour inputs in every firm i, j are assumed to be exogenous to every other firm's TFP error shock in this static model. This is crucial for consistency with the direct own-firm $k_{i,t}, l_{i,t}$ inputs, but also the use of $\mathbf{w}_{ij} k_t, \mathbf{W}^k l_t$ as an *exogenous* effect and, as we will later see, our ability to leverage higher-order spatial lags of the *exogenous* effect as an unconfounded instrument for $w_{ij} y_{j,t}$. Broadly, we can interpret this as meaning factor input decisions are predetermined for this section; capital faces some time-to-build constraint where each firm allocates investment in the previous period based on

the limited information available then, Ω_{t-1} . A similar pattern is assumed of sticky labour decisions, perhaps through job searching and hiring frictions. These restrictive assumptions are relaxed in Dynamic models.

$$K_{i,t} = I_{i,t} + dK_{i,t-1} \quad (11)$$

$$I_{i,t} = \mathbb{E}[I_{i,t}^* | \Omega_{t-1}] \quad (12)$$

$$L_{i,t} = \mathbb{E}[L_{i,t}^* | \Omega_{t-1}] \quad (13)$$

$$K_{i,t}, L_{i,t} \in \Omega_{t-1} \quad (14)$$

$$\mathbb{E}[\varepsilon_{i,t} | \Omega_{t-1}] = 0$$

$$\mathbb{E}[\varepsilon_{i,t} | \begin{pmatrix} \mathbf{k}_t \\ \mathbf{l}_t \end{pmatrix}] = 0_{(2 \times 1)}, \forall i \quad (15)$$

The *endogenous* effect regressor $\sum_{j \neq i} w_{ij} y_{j,t}$ is, unsurprisingly, endogenous, making valid estimation in this setting more complicated. To show the endogeneity requires stacking w_{ij} target-peer weights to an $n \times n$ interactions matrix, $\mathbf{W}$. I invert the *endogenous* component to give a $(\mathbb{I} - \delta_0 \mathbf{W})^{-1}$ coefficient on all variables.

$$\delta_0 < 1 \implies |\mathbb{I} - \delta_0 \mathbf{W}| \neq 0 \quad (16)$$

The inversion is permissible because I can expect a restriction to apply, ruling out iteratively increasing shocks¹⁸. We can interpret this as a ‘spatial lag polynomial’, directly comparable to the lag polynomial and MA inversion procedure in autoregressive time series analysis¹⁹.

$$\mathbf{y}_t = \alpha + \gamma_t \mathbf{l}_t + \beta_0 \mathbf{l}_t + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \delta_0 \mathbf{W} \mathbf{y}_t + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t \quad (17)$$

$$\begin{aligned} &= (\mathbb{I} - \delta_0 \mathbf{W})^{-1} (\alpha + \gamma_t \mathbf{l}_t + \beta_0 \mathbf{l}_t + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t) \\ &= \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k (\alpha + \gamma_t \mathbf{l}_t + \beta_0 \mathbf{l}_t + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t) \end{aligned} \quad (18)$$

$$\begin{aligned} &= \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \alpha + \frac{\gamma_t + \beta_0}{1 - \delta_0} \mathbf{l}_t + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t \\ &\quad + \xi_0 \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^{k+1} \mathbf{k}_t + \zeta_0 \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^{k+1} \mathbf{l}_t + \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \boldsymbol{\varepsilon}_t \end{aligned} \quad (19)$$

$$y_{j,t} = \underbrace{\cdots + \sum_{k=0}^{\infty} [\delta_0^k \mathbf{W}^{k+1}]_{(j, \cdot)} \begin{pmatrix} \mathbf{k}_t & \mathbf{l}_t \end{pmatrix} \begin{pmatrix} \xi_0 \\ \zeta_0 \end{pmatrix}}_{\text{exogenous}_j} + \underbrace{\sum_{k=0}^{\infty} [\delta_0^k \mathbf{W}^k]_{(j, \cdot)} \cdot \boldsymbol{\varepsilon}_t}_{\text{error transformation}_j} \quad (20)$$

It is the primary econometric challenge of this section to overcome the endogeneity of $y_{j,t}$, which is clearly demonstrated in reduced-form equation (20). It contains a linear transformations of all firms’ error shocks $\boldsymbol{\varepsilon}_t$, including $\varepsilon_{i,t}$. This implies:

$$\mathbb{E}[y_{j,t} \varepsilon_{i,t}] \neq 0 \quad (21)$$

¹⁸ $\delta_0 \geq 1$ would imply an exogenous TFP shock to firm k would increase TFP in its peer firm j more than 1x, which would increase TFP in that firm’s peer i more than 1x, and so on until the effect multiplies to infinity, or at least on the knife-edge $\delta_0 = 1$ case, any TFP shock occurring in one firm applies equally and consistently to every firm in a network.

¹⁹ A full algebraic derivation from lines (19) to (20) is available in Appendix A. Note that since the vectors $\mathbf{k}_t, \mathbf{l}_t$ appear twice in (19), as components of $\mathbf{y}_t$ the coefficients can be grouped but it is clearer to separate this into an ‘own-firm’ effect of $\mathbf{k}_t, \mathbf{l}_t$ and a spillover effect of $\mathbf{W}^{k+1} \mathbf{k}_t, \mathbf{W}^{k+1} \mathbf{l}_t \forall k \geq 0$

The ‘Reflection Problem’ poses a second issue as outlined in Manski (1993). Equation (19) shows that $\mathbf{W}y_t$ is a linear combination of all other regressors. In the classical setup, $\mathbf{W}$ is a block diagonal matrix of equal-size groups, and Manski (1993), later reaffirmed in Bramoullé et al. (2009) show that the reduced-form estimation of this model contains fewer estimated parameters than structural parameters, making variation in $y_{j,t}$ indistinguishable to that of all other regressors.

There are two solutions. I could either proceed with an extremely strong exogeneity assumption, (22), which is most likely satisfied by total orthogonality between all contemporaneous idiosyncratic errors hitting peer firms (23). This in turn makes the regressor $y_{j,t}$ exogenous, allowing direct estimation of (7), sidestepping the ‘Reflection Problem’ by assuming away *endogenous* regressors, and having each regressor exogenously identified in itself. For identification, this implies that any productivity shock $\varepsilon_{j,t}$ occurring in peer firm j , if it is contemporaneously correlated with TFP $z_{i,t}$ in target firm i must be attributable to the peer effect coefficient via δ_0 . This precludes any contemporaneous, correlated independent TFP shocks, which might occur on both the global and local level. Model (1.) takes this assumption in 5.2. We should assess these results with the heavy caveat that this strict error orthogonality assumption is unlikely to be credible.

$$\mathbb{E}[\varepsilon_i | \mathbf{W}^k \begin{pmatrix} \alpha & \gamma_t & \beta_0 & \mathbf{k}_t & \mathbf{l}_t \end{pmatrix}] = 0, \forall k$$

$$\mathbb{E}[\varepsilon_{i,t} | \mathbf{W}^k \varepsilon_t] = 0, \forall i, j \quad (22)$$

$$\iff \mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t}] = 0, \forall k \quad (23)$$

Instead, as recommended in Bramoullé et al. (2009), the most defensible approach is to internally instrument the *endogenous* variable, $y_{j,t}$ through deeper spatial lags of the *exogenous* parameters $\mathbf{W}^2 \mathbf{x}_t, \mathbf{W}^3 \mathbf{x}_t \dots \mathbf{W}^k \mathbf{x}_t \forall k > 2$. These are valid instruments under certain testable properties of $\mathbf{W}$ given in Table (I). Essentially, these reduce to checking that the matrices $\mathbf{I}, \mathbf{W}, \mathbf{W}^2$ are linearly independent, which are sufficiently achieved for group interactions by having 2 or more differently-sized groups, or a network diameter of at least 2.

This identification strategy is not a ‘free lunch’. It requires the underlying assumption that the group size, or network diameter, is in itself exogenously determined. This would be intuitively violated, if firms collocate based on the expected productivity benefits of moving to an area, namely to benefit from the supply-side spillovers we aim to investigate. Unfortunately, it is highly likely that firm location decisions are endogenous, this is the whole principle of the New Economic Geography literature whose descriptive patterns of firm movements underlying productivity spillover I here aim to investigate. I am helped here, however, by the use of time-invariant weights. Plausibly, over the observations of a panel dataset up to 20 years, if firm weights apply to each firm on a fixed basis, the temporal variation despite these constant location effects provide credible exogenous variation. On this however, the strategy runs into the limitations of Fama location data discussed in 4.1 The analysis is also helped by the relatively higher number of panel observations to relatively low ratio of confounding movers in the dataset, as well as the numerous, plausibly exogenous reasons why a firm might choose to set up in a location with path dependency, rather than to fully maximise its peer spillover benefits. On the balance of arguments, I take the assumption of network exogeneity and identification approach for models in Tables (XII and IX).

I use the instruments ($\mathbf{W}^2 \mathbf{x}_t$ and other second-order variants) for this section’s group models (Table VI, models (5-8)) and $\mathbf{W}^2 \mathbf{x}_t, \mathbf{W}^3 \mathbf{x}_t, \mathbf{W}^4$ as feasible instruments for this section’s network models (Table VII, model). Equation (24) illustrates the valid 2l moment conditions, noting that I have chosen to display this in a $l \times 2$ matrix for space and conceptual symmetry, as the instruments can be grouped by $\mathbf{k}_t$ and $\mathbf{l}_t$ regressors. The first 2 rows correspond to the exogenous regressors within the reduced-form equation itself, the own-firm classical factors effect from $k_{i,t}, l_{i,t}$ and the *exogenous* peer effect from $k_{j,t}, l_{j,t}$.

The remaining rows are instrument linear combinations of all firm classical factor inputs $\mathbf{k}_t, \mathbf{l}_t$, using the higher-order spatial lags of $\mathbf{W}$ discussed.

$$\mathbb{E}[\epsilon_{i,t} \mid \begin{pmatrix} k_{i,t} & l_{i,t} \\ w_{ij}k_{j,t} & w_{ij}l_{j,t} \\ [\mathbf{W}^2]_{(j,)} \mathbf{k}_t & [\mathbf{W}^2]_{(j,)} \mathbf{l}_t \\ [\mathbf{W}^3]_{(j,)} \mathbf{k}_t & [\mathbf{W}^3]_{(j,)} \mathbf{l}_t \\ \vdots & \vdots \\ [\mathbf{W}^k]_{(j,)} \mathbf{k}_t & [\mathbf{W}^k]_{(j,)} \mathbf{l}_t \end{pmatrix}] = 0_{l \times 2} \quad (24)$$

These instruments and conditions are intuitive: we can see through equation (20) that a linear combination of all firms' contemporaneous capital and labour inputs, $\mathbf{W}^{k+1} \begin{pmatrix} \mathbf{k}_t & \mathbf{l}_t \end{pmatrix} \forall k \geq 0$ are an exogenous component of $y_{j,t}$ with the coefficients ξ_0 and ζ_0 respectively. It has relevance, meaning $\mathbb{E}[y_{j,t} \cdot [\mathbf{W}^k]_{(j,)} \mathbf{x}_t] \neq 0, \forall k > 2$ as long as $\xi_0 \neq 0$ and $\zeta_0 \neq 0$. The relevance conditions themselves are also unsurprising and highly intuitive given the model's setup with GVA equation (7). Note that the relevance conditions are more complex setup in Bramoullé et al. (2009), this is because the factor $\delta_0 \begin{pmatrix} \beta_1 & \beta_2 \end{pmatrix}$ cancels out due the accounting composition of TFP in the *exogenous effect*, which is absent from the general model. This also allows clear understanding of why the TFP-only spillover model (10) fails, as setting both $\xi_0 = \zeta_0 = 0$ means the higher order spatial lags of $\mathbf{k}_t$ and $\mathbf{l}_t$ disappears from the composition of the $y_{j,t}$ term. An Anderson-Rubin test for weak instruments in my nested 3-factor spillover setup can test for the strength of instrument of $\mathbf{k}_t$ and $\mathbf{l}_t$. If I find that they the higher-order spatial lags are jointly weak instruments, that would validate the TFP-only spillover model and suggest $\xi_0 = \zeta_0 = 0$.

Finally, one might observe that $[\mathbf{W}]_{(j,)} \mathbf{x}_t$ is a component of $y_{j,t}$ and so a valid instrument. However, it does not increase the number of instruments I would use in my instrumental specification as it is already included as an exogenous regressor in 17 so does not help increase our identification of $y_{j,t}$.

5.2. LINEAR-IN-MEANS GROUPED MODEL

The first model proceeds with the strict error orthogonality (22) and (23) assumptions. Estimation has the following reduced form:

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \epsilon_{i,t} \quad (\text{from 8})$$

$$+ \delta_0 \sum_{j \neq i} w_{ij} y_{j,t} + (\xi_0 - \delta_0 \beta_1) \cdot \sum_{j \neq i} w_{ij} k_{j,t} + (\zeta_0 - \delta_0 \beta_2) \cdot \sum_{j \neq i} w_{ij} l_{j,t}$$

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \epsilon_{i,t} + \beta_3 \sum_{j \neq i} w_{ij} y_{j,t} + \beta_4 \sum_{j \neq i} w_{ij} k_{j,t} + \beta_5 \sum_{j \neq i} w_{ij} l_{j,t} \quad (25)$$

$$\mathbb{E}[\epsilon_{i,t} \cdot \begin{pmatrix} k_{i,t} & l_{i,t} & y_{j,t} & k_{j,t} & l_{j,t} \end{pmatrix}'] = 0_{(5 \times 1)} \quad (26)$$

$$\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \\ \hat{\beta}_4 \\ \hat{\beta}_5 \end{pmatrix} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \delta_0 \\ \xi_0 - \delta_0 \beta_1 \\ \zeta_0 - \delta_0 \beta_2 \end{pmatrix} \quad (27)$$

With 5 structural and 5 reduced-form parameters, this model is identified. The initial static model applies simple time-invariant weights w_{ij} in a group structure, $\mathbf{W}_{g1}$. As has been indicated by the sum of peers $j \neq i$, I adopt a 'leave-one-out' approach in line with the zero-diagonal structure of $\mathbf{W}_{g1}$ described.

Intuitively, this is the unweighted average of peer TFP within a spatial ring. This weighting matrix is comparable to many sets of weights applied for the 'Linear-in-Means (LMM)' Model common to the peer effects literature, for instance Moffitt and Caspi (2001). I discussed group assignment in section 4.3. The initial Model (1.) estimates (25), meaning a single group $\mathbf{W}_{g1}$ corresponding to the *pc8* group, assuming the full vector of error orthogonality implying regressor exogeneity.

$$\Delta y_{i,t} = \Delta \gamma_t + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} + \Delta \varepsilon_{i,t} + \beta_3 \Delta \sum_{j \neq i} w_{ij} y_{i,t} + \beta_4 \Delta \sum_{j \neq i} w_{ij} k_{j,t} + \beta_5 \Delta \sum_{j \neq i} w_{ij} l_{j,t} \quad (28)$$

$$\mathbb{E}[\epsilon_{i,t} \mid \begin{pmatrix} k_{i,t} & l_{i,t} \\ w_{ij} k_{j,t} & w_{ij} l_{j,t} \\ [\mathbf{W}^2]_{(j,)} \mathbf{k}_t & [\mathbf{W}^2]_{(j,)} \mathbf{l}_t \\ [\mathbf{W}^3]_{(j,)} \mathbf{k}_t & [\mathbf{W}^3]_{(j,)} \mathbf{l}_t \\ \vdots & \vdots \\ [\mathbf{W}^k]_{(j,)} \mathbf{k}_t & [\mathbf{W}^k]_{(j,)} \mathbf{l}_t \end{pmatrix}] = 0_{l \times 2} \quad (\text{from 24})$$

Model (2) instruments these groups via $\mathbf{W}^2 \mathbf{x}$, in line with the moment conditions in (24). As an extension in VIII, I regress the multiple, non-overlapping spatial groups at once. I add $\mathbf{W}_{g1}$, $\mathbf{W}_{g2}$, $\mathbf{W}_{g3}$, concentric doughnut groups for *pc4* and *twa* to run all in a panel OLS regression simultaneously.

I can show algebraically that multiple spatial autoregressors do not change the reflection problem: the moving average inversion's coefficient merely becomes the weighted sum of 3 independent matrices $\theta_g \mathbf{W}_g$. As long as there is sufficient heterogeneity between the weighting matrices, which in these cases is guaranteed by the orthogonality of $\mathbf{W}_g$ matrices to each other, then there is no risk of multicollinearity. Coefficients in the results $\hat{\theta}_1$ for the closest spatial group *pc8* are identical to $\hat{\theta}_0$ in the one-group model. This is an unsurprising result justified by the orthogonality, since the additional regressors do not affect the variation attributable to the $\mathbf{W}_1$ -lagged variables as the regressors with $\mathbf{W}_2$, $\mathbf{W}_3$ are orthogonal.

(Modify equation 18 to include $\sum \mathbf{W}^3$ in the inversion term)

TABLE VII
LMM: 1 peer regressor with group weights,
error orthogonality model vs instrumented

	(1.) <i>y</i>	(2.) <i>y</i>
constant	2.533 (0.044)***	
<i>k</i>	0.261 (0.005)***	0.233 (0.008)***
<i>l</i>	0.650 (0.005)***	0.555 (0.007)***
<i>wg l_y</i>	0.199 (0.004)***	−0.384 (0.177)**
<i>wg l_k</i>	−0.057 (0.003)***	0.118 (0.054)**
<i>wg l_l</i>	−0.118 (0.004)***	0.249 (0.112)**
<i>i</i> fixed effects	✓	
<i>t</i> fixed effects	✓	
Observations	726,283	587,772
R ² (excl. F.E.)	0.3550	0.0012
R ² (Overall)	0.7966	

TABLE VIII
LMM: 3 non-overlapping groups, error
orthogonality vs instrumented specification

	(3.) <i>y</i>	(4.) <i>y</i>
constant	2.342 (0.095)***	
<i>k</i>	0.261 (0.005)***	0.236 (0.011)***
<i>l</i>	0.649 (0.005)***	0.580 (0.020)***
<i>wg l_y</i>	0.198 (0.004)***	−0.715 (0.596)
<i>wg l_k</i>	−0.058 (0.003)***	0.215 (0.180)
<i>wg l_l</i>	−0.117 (0.004)***	0.462 (0.379)
<i>wg2_y</i>	0.019 (0.006)***	4.312 (3.073)
<i>wg2_k</i>	−0.012 (0.004)***	−1.493 (1.088)
<i>wg2_l</i>	−0.001 (0.007)	−2.833 (2.002)
<i>wg3_l</i>	−0.012 (0.019)	26.034 (16.288)
<i>wg3_k</i>	−0.021 (0.012)*	14.842 (9.256)
<i>wg3_y</i>	0.048 (0.017)***	−40.411 (25.266)
<i>i</i> fixed effects	✓	
<i>t</i> fixed effects	✓	
Observations	718,743	581,185

5.3. STATIC DISTANCE DECAY

To improve on the LMM specification, I make a modification where every peer firm is weighted by its influence on the target by the reciprocal of the continuous distance metric calculated in 4.3 $w_{ijt} = f(d)$. This is stylistic of a declining peer influence further away, representing a (lower frequency of social collisions, barriers created via transport and time costs, and an increased difficulty of observation of management practices).

$$z_{it} = \alpha_i + \gamma_t + \beta_0 + \beta_1 \sum_{j \neq i} w_{ijt} z_{jt} + \varepsilon_{it} \quad (29)$$

$$\mathbf{z}_t = (\alpha_i + \gamma_t + \beta_0)\mathbf{1} + \beta_1 \sum_{j \neq i} \Omega_t \mathbf{z}_t + \boldsymbol{\varepsilon}_t \quad (30)$$

$$w_{ijt} \in \Omega_t = \begin{cases} \frac{1}{d_{ij}}, & (1) \text{ gravity} \\ \frac{1}{d_{ij}^2}, & (2) \text{ N.E.G.} \\ \exp -d_{ij} \cdot k & (3) \text{ exponential decline} \end{cases} \quad (31)$$

This represents a likely more accurate declining effect of productivity spillover, rather than the special-case group mean where every firm in an enclosed area, for example, the target's postcode, has an equal influence on the target's TFP. In addition, it allows for better identification following Bramoullé et al. (2009). (explain Bramoullé2009, patterns of) Ω_t

In essence, the LMM specification only contains 3 non-overlapping groups from its location tags. The distance-decayed specification contains as many non-overlapping groups as firms.

This distance-decay specification revises down the estimate from peers to a an elasticity of 0.168, lower than the results given by the LMM specification. This is likely due to (...)
In addition, I introduce controls (...)

TABLE IX
Static panel IV estimation using continuous distance-decay
weighted TFP average of peers

	(1.) y	(2.) y	(3.) y	(4.) y
k	0.271 (0.007)***	0.271 (0.007)***	0.271 (0.007)***	0.267 (0.007)***
l	0.512 (0.006)***	0.512 (0.006)***	0.512 (0.006)***	0.513 (0.006)***
$wd1_k$	0.054 (0.096)			0.250 (0.131)*
$wd1_l$	0.104 (0.184)			0.478 (0.249)*
$wd1_y$	-0.159 (0.286)			-0.740 (0.388)*
$wd2_k$		0.038 (0.080)		
$wd2_l$		0.071 (0.146)		
$wd2_y$		-0.110 (0.233)		
$wd3_k$			0.129 (0.113)	
$wd3_l$			0.230 (0.199)	
$wd3_y$			-0.353 (0.309)	
Observations	854,192	854,192	854,192	854,192
R ²	0.0985	0.1034	0.0980	-0.1213

6. DYNAMIC MODELS

The previous section's static model requires a factor exogeneity assumption outlined in equation (15) for all firms, including the own-firm effect of the production function. In this section, I use a dynamic model to relax this assumption. I also introduce a vector of temporal regressors: an autoregressor of own-firm GVA as a component of its TFP, and allowing for the past peer TFP components I analysed in the static model to influence current-period TFP in a lagged fashion. The structure of this specification now resembles a vector panel, which I will denote with VAR(p,q)²⁰.

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + \sum_{k=1}^p \phi_k y_{i,t-k} + \sum_{k=0}^q \mathbf{z}_{t-k}^{spill} \boldsymbol{\theta}_k + \varepsilon_{i,t} \quad (32)$$

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \varepsilon_{i,t} + \sum_{k=1}^p \phi_k y_{i,t-k} \\ + \sum_{k=0}^q \left[\delta_k \sum_{j \neq i} w_{ij} y_{j,t} + (\xi_k - \delta_k \beta_1) \cdot \sum_{j \neq i} w_{ij} k_{j,t} + (\zeta_k - \delta_k \beta_2) \cdot \sum_{j \neq i} w_{ij} l_{j,t} \right] \quad (33)$$

There are significant theoretical advantages to introducing the q -lags of the spatial spillover. This specification with both spatial and temporal lags accounts for the possibility that the spillover effect from firm j 's production factor has a lag to impact i . Without including the q lags within $z_{i,t}$, if $\mathbf{z}_{t-k}^{spill}$ is correlated with $\mathbf{z}_t$, then estimating δ_0 in a static model will have omitted variable bias (OVB) of the lagged effect from the spillover attributed to the current period.

Adding the autoregressor of $gvaI$ to the specification is highly desirable to a justified decomposition of TFP when including peer effects in the specification. The panel setup with firm fixed effects already accounts for own-firm invariance, and we have shown in the results already given that they account for a significant amount of the variation. In the peer effects setting, the autoregressive lags help control for reverse causality. If $y_{i,t-1}$ is causal for $z_{j,t}$, or feasibly $z_{j,t-1}$ and $\mathbb{E}[z_{j,t-1}, z_{j,t}] \neq 0$, then the presence of $y_{i,t-k}$ as a control helps eliminate that bias. Similarly, theory offers us no clear view on where there could be simultaneity, where variation in $\Delta y_{i,t}$ is driven causally by $\Delta y_{i,t-1}$ and that change in $\Delta y_{i,t}$ in turn drives $\Delta z_{j,t}$. The autoregressive regressor helps control for that. Additionally, the results below suggest the autoregressive parameter coefficient is low, with a log elasticity of 0.005 (significant at the 1% level) in all specifications, suggesting any concerns about reverse causality or simultaneity driven by an AR process are likely minimal. The use of this dynamic regressor improves the plausible exogeneity of $z_{i,j,t}$.

The autoregressive specification requires a return to the one-stage production function with an outcome of GVA, y_{it} . I continue to use the scalar weighted distance TFP metric as the primary regressor of interest z_{jt} . While introducing dynamic lags of

The dynamic panel specification uses a time-varying error orthogonality assumption to find valid identification.

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t}] \quad (34)$$

This might generally be more seen as a more credible assumption than (15) (and we can test for it). This specification can be identified in reduced panel form, which requires first differences to eliminate fixed effects, allowing me to retain the use of firm fixed effects. This follows econometric theory developed by Anderson and Hsiao (1981) and estimator techniques developed in Arellano and Bond (1991) and furthered by Blundell and Bond (1998) which improve efficiency.

²⁰noting that q -lags refers to those on the peer spillover effect, rather than those of a moving average component as is traditional of a VARMA(p,q) specification

6.1. DYNAMIC PANEL IDENTIFICATION

The estimator given in Arellano and Bond (1991) takes the regression equation as given above and allows for endogeneity. Internal instruments of all the endogenous regressors are found, meaning their lagged first differences. Identification is improved in the System GMM approach given in Blundell and Bond (1998) by also including the first lagged, not differenced, lag of the endogenous regressors which meets the usual instrument conditions, as an instrument. It is this System GMM approach that I follow here, so I find the first lagged, not differenced, lag of regressor in x_{it} and its corresponding set of first difference lagged instruments, which in practice just start from $\Delta x_{i,t-j}$ for any j if $x_{i,t}$ is a valid instrument.

I commence finding these valid instruments by assuming the structural specification (7) exhibits no autocorrelation in its error term $\varepsilon_{i,t}$, though I now allow for moderate cross-sectional correlation across firms.

$$\mathbb{E}[\varepsilon_{i,t}\varepsilon_{i,t-1}] = 0, \quad \forall t, t-1 \quad (35)$$

$$\mathbb{E}[x_{i,t}\varepsilon_{i,t}] \neq 0, \quad \exists i \quad (36)$$

$$\mathbb{E}[\varepsilon_{i,t}\varepsilon_{j,t}] \neq 0, \quad \exists i, j \quad (37)$$

$$\mathbb{E}[\varepsilon_t] = 0, \mathbb{E}[\varepsilon_t^2] = \sigma_t^2, \mathbb{E}[\varepsilon_t|\Omega_{t-1}] = 0, \quad \forall i \quad (\text{w.n.}) \quad (38)$$

$$x_{i,t} = \begin{pmatrix} k_{i,t} & l_{i,t} & \sum_{j \neq i} w_{ij,t} z_{j,t} \end{pmatrix}$$

$$\Delta x_{i,t} = \begin{pmatrix} \Delta k_{i,t} & \Delta l_{i,t} & \Delta \sum_{j \neq i} w_{ij,t} z_{j,t} \end{pmatrix}$$

Taking each regressor in turn, I relax the exogeneity assumption of the standard factors in the reduced-form equation, $\Delta k_{i,t}$ and $\Delta l_{i,t}$. I assume a new optimal firm decision on capital based on information, denoted by Ω_t which includes $\varepsilon_{i,t}$ for its respective t . This approach allows the discovery of valid instruments which are exogenous to $\varepsilon_{i,t}$ and $\Delta \varepsilon_{i,t}$. I justify an exclusion restriction for $\Delta k_{i,t-j} \forall j > 1$, as it is predetermined, and $\Delta l_{i,t-k}, k > 2$, as it is flexible.

$$\Delta y_{i,t} = \Delta \gamma_t + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} \Delta z_{j,t} + \Delta \varepsilon_{i,t} \quad (39)$$

$$K_{i,t} = I_{i,t} + dK_{i,t-1} \quad (40)$$

$$I_{i,t} = \mathbb{E}[I_{i,t}^* | \Omega_{t-1}] \quad (41)$$

$$L_{i,t} = \mathbb{E}[L_{i,t}^* | \Omega_t] \quad (42)$$

$$\varepsilon_{i,t} \in \Omega_t$$

$$\implies K_{i,t} \propto \varepsilon_{t-1}$$

$$\implies L_{i,t} \propto \varepsilon_t$$

$$\Delta \varepsilon_{i,t} = \varepsilon_{i,t} - \varepsilon_{i,t-1}$$

$$\Delta k_{i,t} = k_{i,t} - k_{i,t-1} \propto \varepsilon_{i,t-1} \varepsilon_{i,t-2}$$

$$\implies \mathbb{E}[\Delta k_{i,t} \Delta \varepsilon_t] \neq 0$$

$$\implies \mathbb{E}[\Delta k_{i,t-1} \Delta \varepsilon_t] = 0, \mathbb{E}[k_{i,t-1} \varepsilon_t] = 0 \quad (43)$$

$$\Delta l_{i,t} = l_{i,t} - l_{i,t-1} \propto \varepsilon_{i,t}, \varepsilon_{i,t-1}$$

$$\implies \mathbb{E}[\Delta l_{i,t-2} \Delta \varepsilon_t] = 0, \mathbb{E}[l_{i,t-2} \varepsilon_t] = 0 \quad (44)$$

Devising an internal instrument requires a similar evaluation of the components of $y_{j,t}$ as in the

static problem. There is a dynamic corollary to the static derivation of the endogenous term $y_{i,t}$.

$$\begin{aligned} \mathbf{y}_t &= \boldsymbol{\alpha} + \gamma_t \mathbf{t} + \beta_0 \mathbf{t} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \delta_0 \mathbf{W} \mathbf{y}_t + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t \\ &+ \sum_{m=1}^P (\phi_m \mathbb{I} + \delta_m \mathbf{W}) \mathbf{y}_{t-m} + \sum_{m=1}^P (\xi_m - \delta_m \beta_1) \mathbf{W} \mathbf{k}_{t-m} + \sum_{m=1}^P (\zeta_m - \delta_m \beta_2) \mathbf{W} \mathbf{l}_{t-m} \end{aligned} \quad (45)$$

$$\begin{aligned} (\mathbb{I} - \delta_0 \mathbf{W}) \mathbf{y}_t &= \boldsymbol{\alpha} + \gamma_t \mathbf{t} + \beta_0 \mathbf{t} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t \\ &+ \sum_{m=1}^P (\phi_m \mathbb{I} + \delta_m \mathbf{W}) \mathbf{y}_{t-m} \\ &+ \sum_{m=1}^P \mathbf{W} \left[(\xi_m - \delta_m \beta_1) \mathbf{k}_{t-m} + (\zeta_m - \delta_m \beta_2) \mathbf{l}_{t-m} \right] + \boldsymbol{\varepsilon}_t \end{aligned} \quad (46)$$

$$\begin{aligned} \mathbf{y}_t &= \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \left(\boldsymbol{\alpha} + \gamma_t \mathbf{t} + \beta_0 \mathbf{t} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t \right. \\ &+ \sum_{m=1}^P (\phi_m \mathbb{I} + \delta_m \mathbf{W}) \mathbf{y}_{t-m} \\ &\left. + \sum_{m=1}^P \mathbf{W} \left[(\xi_m - \delta_m \beta_1) \mathbf{k}_{t-m} + (\zeta_m - \delta_m \beta_2) \mathbf{l}_{t-m} \right] + \boldsymbol{\varepsilon}_t \right) \end{aligned} \quad (47)$$

$$\begin{aligned} y_{j,t} &= \cdots + \underbrace{\sum_{k=0}^{\infty} [\delta_0^k \mathbf{W}^{k+1}]_{(j, \cdot)} \begin{pmatrix} \mathbf{k}_t & \mathbf{l}_t \end{pmatrix} \begin{pmatrix} \xi_0 \\ \zeta_0 \end{pmatrix}}_{\text{contemporaneous exogenous}_j} \\ &+ \underbrace{\sum_{m=1}^P \sum_{k=0}^{\infty} [\phi_m \delta_0^k \mathbf{W}^k + \delta_m \delta_0^k \mathbf{W}^{k+1}]_{(j, \cdot)} \mathbf{y}_{t-m}}_{\text{entangled historical inertia \& peer effects}_j} \\ &+ \underbrace{\sum_{m=1}^P \sum_{k=0}^{\infty} [\delta_0^k \mathbf{W}^{k+1}]_{(j, \cdot)} \begin{pmatrix} \mathbf{k}_{t-m} & \mathbf{l}_{t-m} \end{pmatrix} \begin{pmatrix} \xi_m - \delta_m \beta_1 \\ \zeta_m - \delta_m \beta_2 \end{pmatrix}}_{\text{historical exogenous network effects}_j} \\ &+ \underbrace{\sum_{k=0}^{\infty} [\delta_0^k \mathbf{W}^k]_{(j, \cdot)} \cdot \boldsymbol{\varepsilon}_t}_{\text{error transformation}_j} \end{aligned} \quad (48)$$

For system GMM, I need to show that the reduced-form first difference regressors are exogenous. The first-order autoregressor, $\Delta \ln(y_{it})$ is endogenous.

$$\Delta y_{i,t} = \Delta \gamma_t + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} + \sum_{k=1}^P \phi_k \Delta y_{i,t-k} + \delta_0 \sum_{j \neq i} w_{ij,t} \Delta z_{j,t} + \Delta \varepsilon_{i,t} \quad (49)$$

$$\Delta \varepsilon_{i,t} = \varepsilon_{i,t} - \varepsilon_{i,t-1} \quad (50)$$

$$\begin{aligned} \Delta y_{i,t-1} &= \cdots + \Delta \varepsilon_{i,t-1} \\ &= \cdots + \varepsilon_{i,t-1} - \varepsilon_{i,t-2} \\ &\implies \mathbb{E}[\Delta y_{i,t-1} \Delta \varepsilon_{i,t}] \neq 0 \\ \Delta y_{i,t-2} &= \cdots + \varepsilon_{i,t-2} - \varepsilon_{i,t-3} \\ &\implies \mathbb{E}[\Delta y_{i,t-1} \Delta \varepsilon_{i,t}] = 0 \end{aligned}$$

This is confirmed by the Anderson-Rubin (AR(1)) test in my results, which rejects the null hypothesis at the 1% level for all models, suggesting that first-order lags of log GVA are a weak instrument (Why? Shouldn't they fail exclusion restriction, not instrument correlation). In addition, I assume that capital, $k_{i,t}$ is pre-determined, as investment is chosen one period ahead, in line with standard Perpetual Inventory Method approaches²¹ (Review predetermined material. Additionally, explain the logic of instrumenting for capital)

To overcome this endogeneity, I introduce deeper autoregressive lags as instrumental variables, as proposed by Anderson and Hsiao (1981). I use 3 autoregressive instrument lags starting from $t - 2$: $\ln(y_{i,t-2}), \ln(y_{i,t-2}), \ln(y_{i,t-p-3})$ for log GVA, and 2 lags for capital $\ln(k_{i,t-p-1}), \ln(k_{t-p-2})$. An Anderson-Rubin (AR(2)) test shows that deep lags are a valid instrument for log GVA. The test applied to the second lag of $\ln(gva_1)$ fails to reject the null hypothesis of for any specification, so I assume it is a valid instrument.

Additionally, using multiple deep lags allows for overidentification: in this basic AR(1) specification, I have $k = 4$ regressors and $l = 7$ instruments. I could increase this. Fame is rich on the time dimension, with up to 19 lags at most for a firm, and a median of 6 lags, so I could safely provide more lags of regressors as instruments. Further lags is likely to have a decreasing marginal effect on explanatory power and additionally (O...)

(Discuss introducing a new weighting method using past firm similarity.)

I estimate the equation fulfilling the vector of moment conditions, (51) via GMM.

$$\mathbb{E} \left[\Delta \varepsilon_{it} \cdot \begin{pmatrix} \Delta \ln(y_{i,t-2}) \\ \Delta \ln(y_{i,t-3}) \\ \Delta \ln(y_{i,t-4}) \\ \Delta \ln(k_{i,t-2}) \\ \Delta \ln(k_{i,t-3}) \\ \Delta \sum_j w_{ijt} z_{jt} \end{pmatrix} \right] = 0 \quad (51)$$

(I should introduce a control (basically a diff-in-diff?) of peers from the same industry across the country (outside the spatial bubble).)

Finally, a Sargan-Hansen J-test is implemented to check if these instruments are over-identified, meaning the moment conditions cannot be fulfilled with 95% confidence. In all models, the J-test fails to reject the null hypothesis, meaning the model's moment conditions are jointly correct.

(!!!!) Following above reasoning, lags of $\Delta \sum_{j \neq i} w_{i,j,t} z_{j,t}$ of a higher order than 2 are exogenous to my reduced-form specification. However,

6.2. INSTRUMENTAL IDENTIFICATION UNDER ERROR AUTOCORRELATION

I appear to have strong identification given assumptions (35) and (37). The instruments above exhibit strong correlation under Anderson-Rubin tests and exclusion restriction has been derived. However, there exists a niche case of diagonally-correlated errors, namely despite (35) and (37), I ought to consider the following:

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t-k}] \neq 0, \quad \exists i, j, k \quad (52)$$

This is not as implausible as it may appear due to this analysis' peer effects setting. There may be some local omitted shock v_t , that impacts peer firm A's TFP via ε_t and impacts target firm B's TFP after a lag via ε_{t-k} . This could be a price shock with random adjustment (or)

²¹Noting that this is ordinarily done on a quarterly basis, however, due to data limitations we adopt the same thing here even though Fame provides observations annually.

TABLE X
Dynamic panel: baseline instrumental specification

	(1.) y	(2.) y	(3.) y
constant	−0.864 (0.697)	−0.114 (0.869)	−0.673 (0.793)
y_{t-1}	−0.000 (0.017)	0.000 (0.019)	−0.001 (0.019)
k	0.431 (0.070)***	0.363 (0.080)***	0.367 (0.085)***
l	0.597 (0.112)***	0.508 (0.122)***	0.696 (0.152)***
wdl_y	−0.108 (0.383)	−0.293 (0.414)	0.659 (0.780)
$wdl_{y_{t-1}}$	0.783 (0.243)**	0.819 (0.252)**	0.333 (0.476)
wdl_k	0.048 (0.288)	0.226 (0.286)	−0.469 (0.428)
$wdl_{k,t-1}$	−0.286 (0.165)	−0.322 (0.160)*	0.029 (0.259)
wdl_l	0.527 (0.366)	0.478 (0.351)	0.179 (0.638)
$wdl_{l_{t-1}}$	−0.722 (0.237)**	−0.632 (0.234)**	−0.554 (0.397)
y_{t-2}		0.003 (0.002)*	
Hansen Test	X (0.644)	X (0.864)	X (0.844)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	✓ (0.033)**	X (0.051)*	X (0.209)
Instruments	45	45	36
i fixed effects	✓	✓	✓
t fixed effects	✓	✓	✓
Observations	686,164	606,795	686,164

Hansen J-test for over-identification: ‘X’ indicates insufficient evidence at the 95% level to reject the null-hypothesis that the model is identified.

Anderson-Rubin AR(1) test: ‘✓’ indicates the use of single-order lags is a weak instrument, as expected from a first-differencing SystemGMM approach.

Anderson-Rubin AR(2) test: ‘X’ indicates insufficient evidence to reject the use of second-order lags due to their being a weak instrument, as expected from the first-differencing with lags greater than 2nd-order to overcome contemporaneous endogeneity. ‘checkmark’ indicates some instruments used were weak.

TABLE XI
Dynamic panel: deeper vector regressor lags

	(1.) <i>y</i>	(2.) <i>y</i>	(3.) <i>y</i>	(4.) <i>y</i>	(5.) <i>y</i>
constant	−0.257 (0.299)	−0.307 (0.791)	−0.812 (0.736)	−1.765 (1.034)	−1.288 (0.993)
$y_t - 1$	0.002 (0.002)	0.002 (0.002)	0.001 (0.002)	0.004 (0.002)*	0.004 (0.002)
k	0.324 (0.043)***	0.353 (0.041)***	0.330 (0.051)***	0.190 (0.088)*	0.202 (0.078)**
l	0.694 (0.070)***	0.678 (0.064)***	0.685 (0.073)***	0.669 (0.123)***	0.730 (0.102)***
wdl_y	1.118 (0.094)***	1.007 (0.222)***	1.105 (0.231)***	0.832 (0.308)**	0.804 (0.342)*
wdl_k	−0.399 (0.065)***	−0.344 (0.100)***	−0.328 (0.169)	0.197 (0.324)	0.172 (0.300)
wdl_l	−0.707 (0.089)***	−0.641 (0.137)***	−0.695 (0.169)***	−0.686 (0.344)*	−0.792 (0.364)*
wdl_y_{t-1}		0.003 (0.008)	−0.001 (0.011)	−0.001 (0.016)	0.001 (0.018)
$wdl_{k,t-1}$		−0.004 (0.004)	−0.005 (0.009)	−0.036 (0.020)	−0.032 (0.018)
wdl_l_{t-1}		−0.002 (0.006)	0.001 (0.010)	0.015 (0.025)	0.016 (0.026)
wdl_y_{t-2}			−0.001 (0.005)	0.000 (0.008)	−0.004 (0.010)
$wdl_{k,t-2}$			−0.001 (0.004)	−0.016 (0.009)	−0.011 (0.009)
wdl_l_{t-2}			0.000 (0.005)	0.007 (0.012)	0.009 (0.013)
wdl_y_{t-3}				−0.002 (0.005)	−0.002 (0.006)
$wdl_{k,t-3}$				−0.006 (0.004)	−0.004 (0.004)
wdl_l_{t-3}				0.004 (0.006)	0.002 (0.007)
wdl_y_{t-4}					0.002 (0.005)
$wdl_{k,t-4}$					−0.001 (0.003)
wdl_l_{t-4}					−0.002 (0.005)
Hansen Test	X (0.678)	✓ (0.010)**	✓ (0.007)***	✓ (0.042)**	✓ (0.001)***
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.538)	X (0.498)	X (0.757)	X (0.307)	X (0.362)
Instruments	42	49	48	46	46
i fixed effects	✓	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓	✓
Observations	717,300	717,300	606,795	514,786	436,247

Hansen J-test for over-identification: 'X' indicates insufficient evidence at the 95% level to reject the null-hypothesis that the model is identified. Anderson-Rubin AR(1) test: Model includes a constant as the baseline for t fixed effects.

TABLE XII
Dynamic panel: deeper autoregressive lags

	(1.) y	(2.) y	(3.) y	(4.) y
constant	-0.257 (0.299)	-0.752 (1.173)	-0.667 (1.216)	0.261 (0.895)
y_{t-1}	0.002 (0.002)	0.008 (0.003)*	0.008 (0.004)*	0.016 (0.004)***
k	0.324 (0.043)***	0.171 (0.086)*	0.172 (0.086)*	0.199 (0.073)**
l	0.694 (0.070)***	0.685 (0.125)***	0.677 (0.128)***	0.568 (0.099)***
wdl_y	1.118 (0.094)***	0.488 (0.339)	0.473 (0.344)	0.470 (0.369)
wdl_k	-0.399 (0.065)***	0.386 (0.308)	0.383 (0.306)	0.164 (0.284)
wdl_l	-0.707 (0.089)***	-0.667 (0.361)	-0.651 (0.365)	-0.434 (0.305)
y_{t-2}		0.003 (0.002)	0.003 (0.002)	0.007 (0.002)***
$wdl_{y_{t-1}}$		0.014 (0.018)	0.015 (0.019)	0.022 (0.018)
$wdl_{y_{t-2}}$		0.004 (0.009)	0.005 (0.009)	0.005 (0.009)
$wdl_{y_{t-3}}$		-0.000 (0.005)	-0.000 (0.005)	0.001 (0.005)
$wdl_{k,t-1}$		-0.042 (0.019)*	-0.042 (0.019)*	-0.022 (0.016)
$wdl_{k,t-2}$		-0.018 (0.009)	-0.018 (0.009)	-0.007 (0.008)
$wdl_{k,t-3}$		-0.007 (0.004)	-0.007 (0.004)	-0.003 (0.004)
$wdl_{l_{t-1}}$		0.018 (0.025)	0.017 (0.025)	-0.002 (0.020)
$wdl_{l_{t-2}}$		0.008 (0.012)	0.008 (0.012)	0.001 (0.010)
$wdl_{l_{t-3}}$		0.005 (0.006)	0.005 (0.006)	-0.002 (0.005)
y_{t-3}			0.000 (0.001)	0.003 (0.001)*
y_{t-4}				0.004 (0.001)**
Hansen Test	X (0.678)	✓ (0.020)**	✓ (0.012)**	X (0.106)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.538)	X (0.755)	X (0.733)	X (0.976)
Instruments	42	47	47	46
i fixed effects	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓
Observations	717,300	514,786	514,786	436,247

Model includes a constant as the baseline for t fixed effects.

Additionally, relaxing (35) to allow for weak error autocorrelation between limited $\varepsilon_{i,t}$ and $\varepsilon_{i,t-k}$ for limited k proves a similar problem.

(Firstly, I test for error autocorrelation.)

7. EXTENSIONS

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A. APPENDIX 1: IDENTIFICATION UNDER DIFFERING SPILLOVER ASSUMPTIONS

A.1. IDENTIFICATION FULL DERIVATION UNDER THE 3-FACTOR SPILLOVER MODEL

$$\begin{aligned}
y_t &= \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \left(\alpha + (\gamma_t + \beta_0) \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t \right. \\
&\quad \left. + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t \right) \\
&= \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k (\gamma_t + \beta_0) \boldsymbol{\iota} + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \alpha + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \boldsymbol{\varepsilon}_t \\
&\quad + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \beta_1 \mathbf{k}_t + (\xi_0 - \delta_0 \beta_1) \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{k}_t \\
&\quad + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \beta_2 \mathbf{l}_t + (\zeta_0 - \delta_0 \beta_2) \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{l}_t \\
&= \frac{\gamma_t + \beta_0}{1 - \theta_0} \boldsymbol{\iota} + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \alpha + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \boldsymbol{\varepsilon}_t \\
&\quad + \beta_1 \mathbf{k}_t + \sum_{k=1}^{\infty} \theta_0^k \mathbf{W}^k \beta_1 \mathbf{k}_t + (\xi_0 - \delta_0 \beta_1) \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{k}_t \\
&\quad + \beta_2 \mathbf{l}_t + \sum_{k=1}^{\infty} \theta_0^k \mathbf{W}^k \beta_2 \mathbf{l}_t + (\zeta_0 - \delta_0 \beta_2) \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{l}_t \\
&= \frac{\gamma_t + \beta_0}{1 - \theta_0} \boldsymbol{\iota} + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \alpha + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \boldsymbol{\varepsilon}_t \\
&\quad + \beta_1 \mathbf{k}_t + \theta_0 \beta_1 \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{k}_t + (\xi_0 - \delta_0 \beta_1) \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{k}_t \\
&\quad + \beta_2 \mathbf{l}_t + \theta_0 \beta_2 \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{l}_t + (\zeta_0 - \delta_0 \beta_2) \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{l}_t \\
&= \frac{\gamma_t + \beta_0}{1 - \theta_0} \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \alpha + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \boldsymbol{\varepsilon}_t \\
&\quad + (\xi_0 - \delta_0 \beta_1 + \theta_0 \beta_1) \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{k}_t \\
&\quad + (\zeta_0 - \delta_0 \beta_2 + \theta_0 \beta_2) \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{l}_t
\end{aligned}$$

A.2. NON-IDENTIFICATION UNDER THE TFP-ONLY SPILLOVER MODEL

$$\begin{aligned}
\mathbf{y}_t &= \boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \theta_0 \mathbf{W} \mathbf{y}_t + \xi_0 \mathbf{W} \cdot (\beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t) + \boldsymbol{\varepsilon}_t \\
&= (\mathbb{I} - \theta_0 \mathbf{W})^{-1} (\boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \xi_0 \mathbf{W} \cdot (\beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t) + \boldsymbol{\varepsilon}_t) \\
&= \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \cdot (\boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \xi_0 \mathbf{W} \cdot (\beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t) + \boldsymbol{\varepsilon}_t) \\
&= \frac{\gamma_t + \beta_0}{1 - \theta_0} \boldsymbol{\iota} + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \boldsymbol{\alpha} + \boldsymbol{\beta} \mathbf{x}_t + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \boldsymbol{\beta} \cdot (\theta_0 + \xi_0) + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \boldsymbol{\varepsilon}_t \\
y_{j,t} &= \cdots + \underbrace{\sum_{k=0}^{\infty} [\theta_0^k \mathbf{W}^{k+1}]_{(j, \cdot)} \begin{pmatrix} \mathbf{k}_t & \mathbf{l}_t \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}}_{\text{exogenous}_j} \cdot (\theta_0 + \xi_0) + \underbrace{\sum_{k=0}^{\infty} [\theta_0^k \mathbf{W}^k]_{(j, \cdot)} \cdot \boldsymbol{\varepsilon}_t}_{\text{error transformation}_j} \\
&= \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k (\mathbf{x}_t \boldsymbol{\beta} + \mathbf{W} \mathbf{x}_t \boldsymbol{\beta} \cdot \xi_0) \\
&= \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \mathbf{x}_t \boldsymbol{\beta} + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \boldsymbol{\beta} \cdot \xi_0 \\
&= \mathbf{x}_t \boldsymbol{\beta} + \sum_{k=1}^{\infty} \theta_0^k \mathbf{W}^k \mathbf{x}_t \boldsymbol{\beta} + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \boldsymbol{\beta} \cdot \xi_0 \\
&= \mathbf{x}_t \boldsymbol{\beta} + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \boldsymbol{\beta} \cdot \theta_0 + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \boldsymbol{\beta} \cdot \xi_0 \\
&= \mathbf{x}_t \boldsymbol{\beta} + \left(\sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \right) \boldsymbol{\beta} \cdot (\theta_0 + \xi_0) \\
\mathbf{y}_t &= \frac{\alpha}{1 - \theta_0} \boldsymbol{\iota} + \mathbf{x}_t \boldsymbol{\beta} + \left(\sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \right) \boldsymbol{\beta} \cdot (\theta_0 + \xi_0) + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \boldsymbol{\varepsilon}_t
\end{aligned}$$

B. APPENDIX 2

TABLE XIII
Dynamic panel: higher-order autoregressive lags

	(1)	(2)	(3)	(4)
	$\ln(gva_1)$	$\ln(gva_1)$	$\ln(gva_1)$	$\ln(gva_1)$
constant	1.371 (0.134)***	1.446 (0.843)	0.208 (2.578)	0.459 (2.844)
$\ln(gva_1)_{t-1}$	0.005 (0.001)***	0.019 (0.104)	0.128 (0.295)	-0.020 (0.256)
$\ln(gva_1)_{t-2}$		0.002 (0.003)	0.004 (0.101)	0.094 (0.300)
$\ln(gva_1)_{t-3}$			-0.001 (0.006)	0.036 (0.162)
$\ln(gva_1)_{t-4}$				-0.003 (0.004)
$\ln(employees)$	0.696 (0.038)***	0.762 (0.097)***	0.736 (0.149)***	0.659 (0.323)*
$\ln(total_assets)$	0.259 (0.041)***	0.190 (0.090)*	0.248 (0.121)*	0.291 (0.305)
tfp_wav_1	0.503 (0.028)***	0.548 (0.059)***	0.508 (0.076)***	0.466 (0.184)*
Hansen Test	X (0.417)	X (0.922)	X (0.881)	X (0.826)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	X (0.140)	X (0.212)
AR(2) Test	X (0.746)	X (0.925)	X (0.802)	X (0.668)
i fixed effects	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓
Observations	713,339	603,058	511,396	433,185

APPENDIX 3

TABLE XIV
Dynamic panel: higher-order regressor lags 1-5

	(1) ln(<i>gva</i> ₁)	(2) ln(<i>gva</i> ₁)	(3) ln(<i>gva</i> ₁)	(4) ln(<i>gva</i> ₁)	(5) ln(<i>gva</i> ₁)
constant	1.124 (0.113)***	0.956 (0.101)***	0.743 (0.093)***	0.584 (0.091)***	0.543 (0.094)***
ln(<i>gva</i> ₁) _{<i>t</i> - 1}	0.002 (0.001)*	0.003 (0.001)***	0.005 (0.001)***	0.005 (0.001)***	0.005 (0.001)***
<i>tfp_wav</i> _{1<i>t</i> - 1}	0.066 (0.006)***	0.077 (0.007)***	0.069 (0.007)***	0.062 (0.007)***	0.061 (0.008)***
<i>tfp_wav</i> _{1<i>t</i> - 2}		0.052 (0.005)***	0.055 (0.006)***	0.050 (0.007)***	0.051 (0.007)***
<i>tfp_wav</i> _{1<i>t</i> - 3}			0.043 (0.005)***	0.048 (0.006)***	0.049 (0.006)***
<i>tfp_wav</i> _{1<i>t</i> - 4}				0.030 (0.005)***	0.038 (0.006)***
<i>tfp_wav</i> _{1<i>t</i> - 5}					0.028 (0.005)***
ln(<i>employees</i>)	0.686 (0.036)***	0.695 (0.038)***	0.670 (0.040)***	0.644 (0.044)***	0.665 (0.052)***
ln(<i>total_assets</i>)	0.270 (0.040)***	0.260 (0.042)***	0.287 (0.045)***	0.317 (0.049)***	0.294 (0.057)***
<i>tfp_wav</i> ₁	0.508 (0.028)***	0.514 (0.029)***	0.492 (0.030)***	0.469 (0.033)***	0.480 (0.038)***
Hansen Test	X (0.492)	X (0.968)	X (0.453)	X (0.084)*	X (0.080)*
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.279)	X (0.612)	X (0.801)	X (0.854)	X (0.729)
<i>i</i> fixed effects	✓	✓	✓	✓	✓
<i>t</i> fixed effects	✓	✓	✓	✓	✓
Observations	713,339	603,058	511,396	433,185	365,908

TABLE XV
Dynamic panel: higher-order regressor lags 6-10

	(1) ln(<i>gva</i> ₁)	(2) ln(<i>gva</i> ₁)	(3) ln(<i>gva</i> ₁)	(4) ln(<i>gva</i> ₁)	(5) ln(<i>gva</i> ₁)
constant	0.421 (0.082)***	0.388 (0.081)***	0.270 (0.077)***	0.199 (0.083)*	0.116 (0.093)
ln(<i>gva</i> ₁) _{<i>t</i> - 1}	0.004 (0.001)***	0.004 (0.001)**	0.003 (0.001)*	0.002 (0.002)	0.002 (0.002)
<i>tfp_wav</i> _{1<i>t</i> - 1}	0.055 (0.008)***	0.056 (0.008)***	0.047 (0.008)***	0.042 (0.009)***	0.036 (0.011)**
<i>tfp_wav</i> _{1<i>t</i> - 2}	0.045 (0.007)***	0.045 (0.007)***	0.040 (0.007)***	0.037 (0.007)***	0.035 (0.009)***
<i>tfp_wav</i> _{1<i>t</i> - 3}	0.046 (0.006)***	0.046 (0.006)***	0.039 (0.006)***	0.033 (0.007)***	0.029 (0.008)***
<i>tfp_wav</i> _{1<i>t</i> - 4}	0.034 (0.006)***	0.031 (0.006)***	0.024 (0.006)***	0.019 (0.007)**	0.017 (0.008)*
<i>tfp_wav</i> _{1<i>t</i> - 5}	0.029 (0.005)***	0.030 (0.006)***	0.023 (0.006)***	0.019 (0.006)**	0.017 (0.006)**
<i>tfp_wav</i> _{1<i>t</i> - 6}	0.018 (0.004)***	0.024 (0.005)***	0.021 (0.005)***	0.019 (0.005)***	0.017 (0.006)**
<i>tfp_wav</i> _{1<i>t</i> - 7}		0.019 (0.005)***	0.017 (0.005)***	0.015 (0.006)**	0.011 (0.007)
<i>tfp_wav</i> _{1<i>t</i> - 8}			0.013 (0.004)**	0.016 (0.005)**	0.017 (0.006)**
<i>tfp_wav</i> _{1<i>t</i> - 9}				0.010 (0.005)*	0.013 (0.005)*
ln(<i>employees</i>)	0.628 (0.049)***	0.636 (0.053)***	0.582 (0.055)***	0.548 (0.061)***	0.523 (0.073)***
<i>tfp_wav</i> _{1<i>t</i> - 10}					0.016 (0.006)**
ln(<i>total_assets</i>)	0.334 (0.054)***	0.325 (0.058)***	0.384 (0.061)***	0.421 (0.067)***	0.448 (0.080)***
<i>tfp_wav</i> ₁	0.452 (0.036)***	0.457 (0.038)***	0.420 (0.040)***	0.397 (0.044)***	0.380 (0.052)***
Hansen Test	X (0.097)*	X (0.122)	X (0.450)	X (0.637)	X (0.435)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.924)	X (0.962)	X (0.568)	X (0.507)	X (0.450)
<i>i</i> fixed effects	✓	✓	✓	✓	✓
<i>t</i> fixed effects	✓	✓	✓	✓	✓
Observations	307,394	256,016	210,798	171,125	136,527

TABLE XVI
Dynamic panel: higher-order VAR(2,4) structures

	(1) ln(<i>gva</i> ₁)	(2) ln(<i>gva</i> ₁)	(3) ln(<i>gva</i> ₁)	(4) ln(<i>gva</i> ₁)	(5) ln(<i>gva</i> ₁)	(6) ln(<i>gva</i> ₁)
constant	1.164 (0.573)*	0.498 (1.561)	0.398 (1.667)	1.494 (0.428)***	0.333 (1.469)	1.351 (0.307)***
ln(<i>gva</i> ₁) _{<i>t</i>−1}	−0.011 (0.117)	0.081 (0.280)	−0.028 (0.251)	−0.129 (0.081)	0.087 (0.276)	−0.125 (0.063)*
ln(<i>gva</i> ₁) _{<i>t</i>−2}	−0.000 (0.003)	0.006 (0.111)	0.076 (0.385)	0.004 (0.002)	−0.001 (0.111)	0.006 (0.002)**
ln(<i>gva</i> ₁) _{<i>t</i>−3}		−0.001 (0.005)	0.034 (0.156)		−0.002 (0.004)	
ln(<i>gva</i> ₁) _{<i>t</i>−4}			−0.002 (0.004)			
<i>tfp</i> _{wav1}	0.551 (0.052)***	0.504 (0.078)***	0.475 (0.201)*	0.525 (0.049)***	0.488 (0.073)***	0.524 (0.050)***
<i>tfp</i> _{wav1t−1}	0.099 (0.102)	0.005 (0.255)	0.082 (0.217)	0.196 (0.073)**	−0.005 (0.249)	0.189 (0.055)***
<i>tfp</i> _{wav1t−2}	0.060 (0.011)***	0.036 (0.099)	−0.019 (0.342)	0.072 (0.012)***	0.046 (0.102)	0.070 (0.010)***
<i>tfp</i> _{wav1t−3}				0.056 (0.009)***	0.039 (0.020)	0.065 (0.008)***
<i>tfp</i> _{wav1t−4}						0.042 (0.006)***
ln(<i>employees</i>)	0.737 (0.099)***	0.720 (0.146)***	0.643 (0.382)	0.650 (0.075)***	0.705 (0.148)***	0.657 (0.079)***
ln(<i>total_assets</i>)	0.209 (0.088)*	0.253 (0.119)*	0.307 (0.367)	0.272 (0.074)***	0.271 (0.119)*	0.266 (0.079)***
Hansen Test	X (0.971)	X (0.855)	X (0.849)	X (0.999)	X (0.851)	X (0.232)
AR(1) Test	✓ (0.002)***	X (0.124)	X (0.163)	✓ (0.000)***	X (0.111)	✓ (0.000)***
AR(2) Test	X (0.889)	X (0.878)	X (0.775)	X (0.092)*	X (0.837)	✓ (0.030)**
<i>i</i> fixed effects	✓	✓	✓	✓	✓	✓
<i>t</i> fixed effects	✓	✓	✓	✓	✓	✓
Observations	603,058	511,396	433,185	511,396	511,396	433,185